Model Comparison Report for Last mRS Prediction

This report compares multiple machine learning models for predicting the Last mRS outcome, using radiomic features alone and combined with clinical features.

Feature Sets:
- Radiomics: Only radiomic features (quantitative features from images)
- Radiomics+Clinical: Radiomic plus clinical features (e.g., age, sex, clinical scores)

Tasks:
- Regression: Predicting Last mRS as a continuous value (e.g., 2.5)
- Classification: Predicting Last mRS as a discrete class (e.g., 0, 1, 2, 3, 4, 5, 6)

Metrics:
- MAE: Mean Absolute Error (lower is better, regression)
- R^2: R-squared (higher is better, regression)
- Accuracy: Fraction of correct predictions (classification)
- F1, Precision, Recall: Standard classification metrics (higher is better)

Key Results:
- Tree-based models (CatBoost, RandomForest, XGBoost, LightGBM) perform best, often with perfect or near-perfect scores.
- Linear/Logistic regression performs worse, indicating non-linear relationships.
- Adding clinical features does not significantly improve performance for tree-based models.
- Perfect scores may indicate overfitting; results should be validated on more data.

Limitations & Next Steps:
- Results may not generalize due to small sample size and possible overfitting.
- Use cross-validation and/or external validation for more robust assessment.
- Consider feature importance analysis and model interpretability tools.

Radiomics - Classification Model Comparison

Model	MAE	R^2	Accuracy	F1	Precision	Recall
CatBoost	nan	nan	1.0	1.0	1.0	1.0
RandomForest	nan	nan	0.964	0.943	0.982	0.929
XGBoost	nan	nan	1.0	1.0	1.0	1.0
LightGBM	nan	nan	1.0	1.0	1.0	1.0
Logistic	nan	nan	0.143	0.075	0.043	0.286

Radiomics - Regression Model Comparison

Model	MAE	R^2	Accuracy	F1	Precision	Recall
CatBoost	0.285	0.938	nan	nan	nan	nan
RandomForest	0.198	0.976	nan	nan	nan	nan
XGBoost	0.0	1.0	nan	nan	nan	nan
LightGBM	0.286	0.928	nan	nan	nan	nan
Linear	4.903	-42.272	nan	nan	nan	nan

Radiomics+clinical - Classification Model Comparison

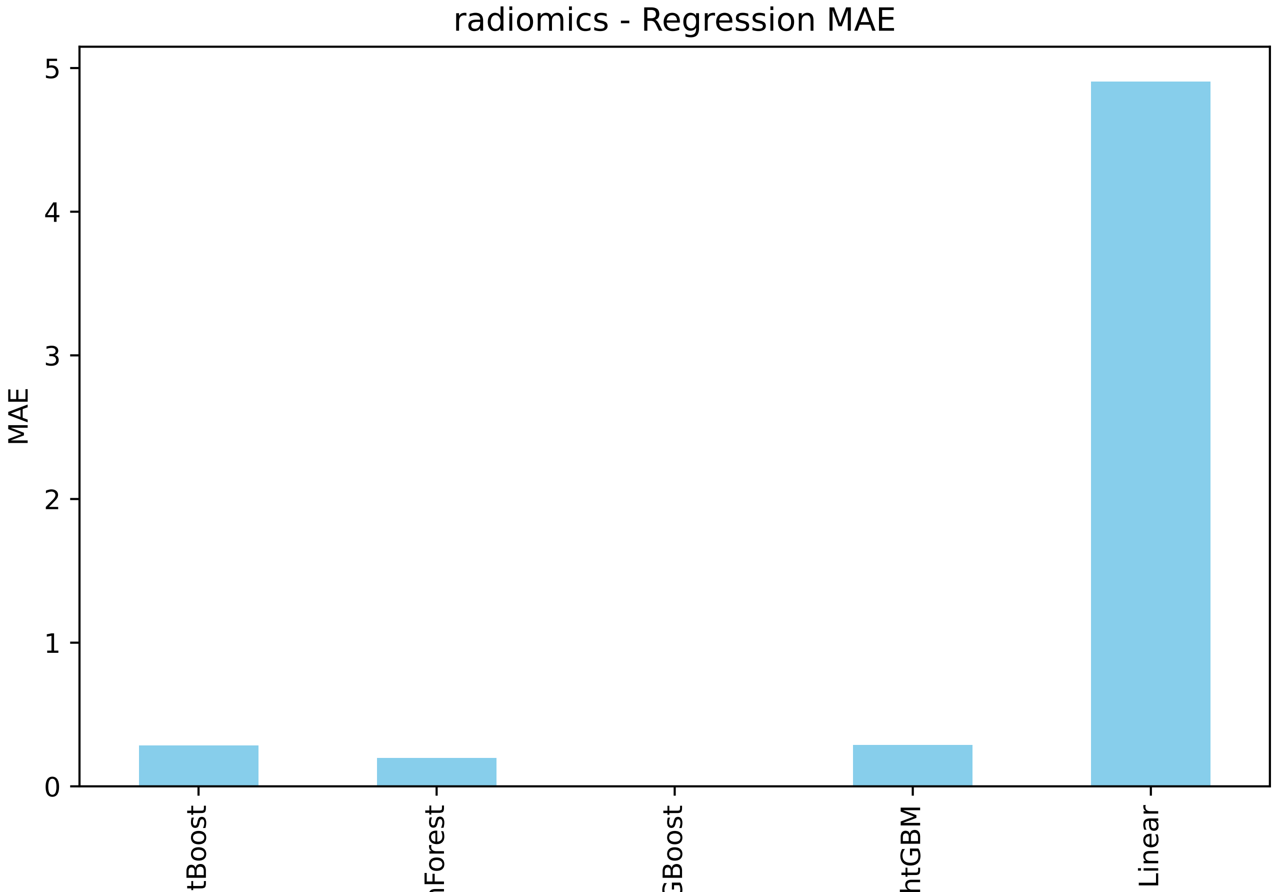
Model	MAE	R²	Accuracy	F1	Precision	Recall
CatBoost	nan	nan	1.0	1.0	1.0	1.0
RandomForest	nan	nan	1.0	1.0	1.0	1.0
XGBoost	nan	nan	1.0	1.0	1.0	1.0
LightGBM	nan	nan	1.0	1.0	1.0	1.0
Logistic	nan	nan	0.321	0.153	0.14	0.189

Radiomics+clinical - Regression Model Comparison

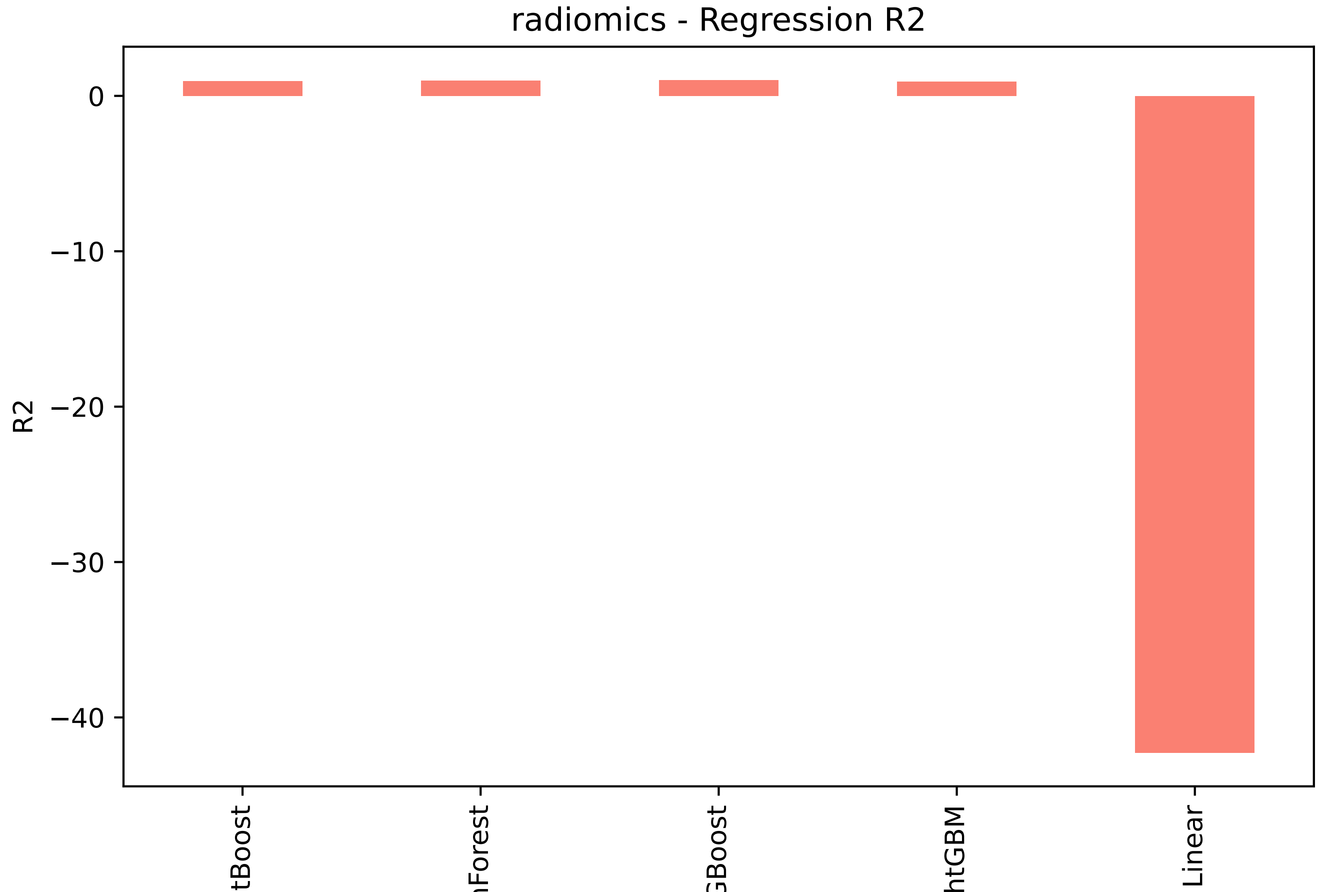
Model	MAE	R^2	Accuracy	F1	Precision	Recall
CatBoost	0.173	0.96	nan	nan	nan	nan
RandomForest	0.093	0.989	nan	nan	nan	nan
XGBoost	0.0	1.0	nan	nan	nan	nan
LightGBM	0.083	0.99	nan	nan	nan	nan
Linear	0.0	1.0	nan	nan	nan	nan

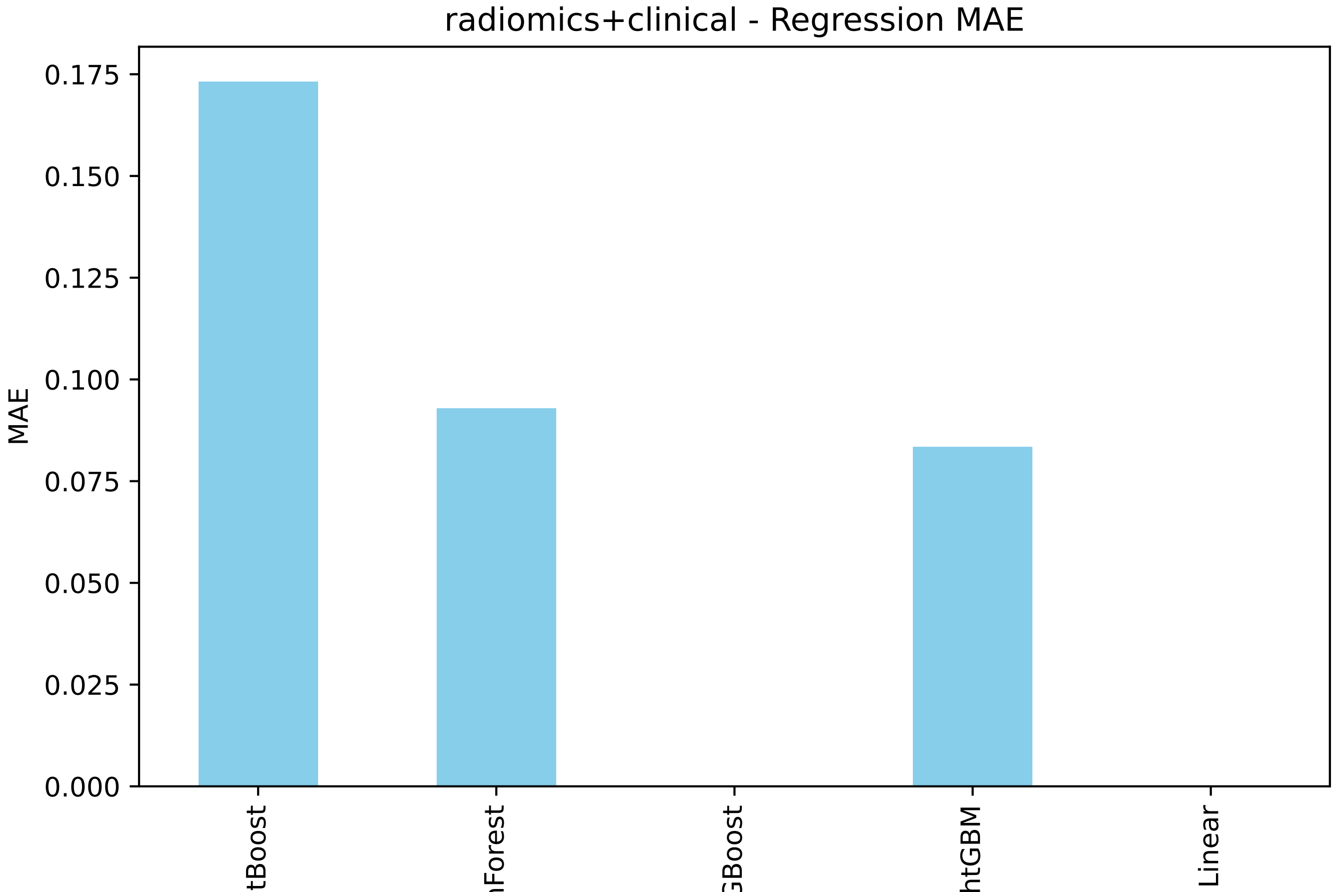
Limitations & Recommendations

- The models may be overfitting, as indicated by perfect or near-perfect scores.
- The dataset is relatively small; results may not generalize to new data.
- Cross-validation or external validation is recommended.
- Consider exploring feature importance and interpretability tools (e.g., SHAP, permutation importance).
- Further work: test on larger, independent datasets; explore model calibration; report confidence intervals.

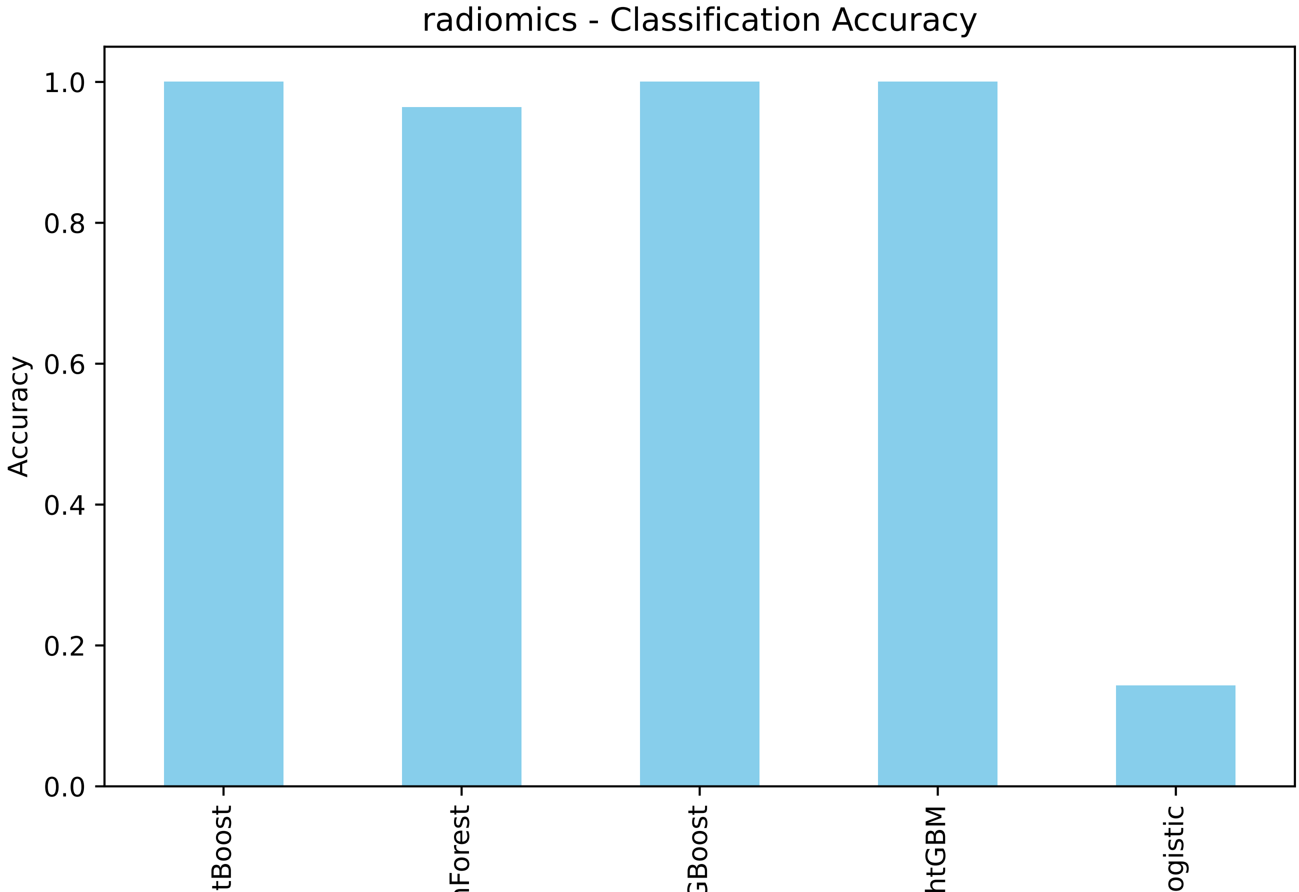


radiomics - Regression MAE
MAE
0
1
2
3
4
5
Linear

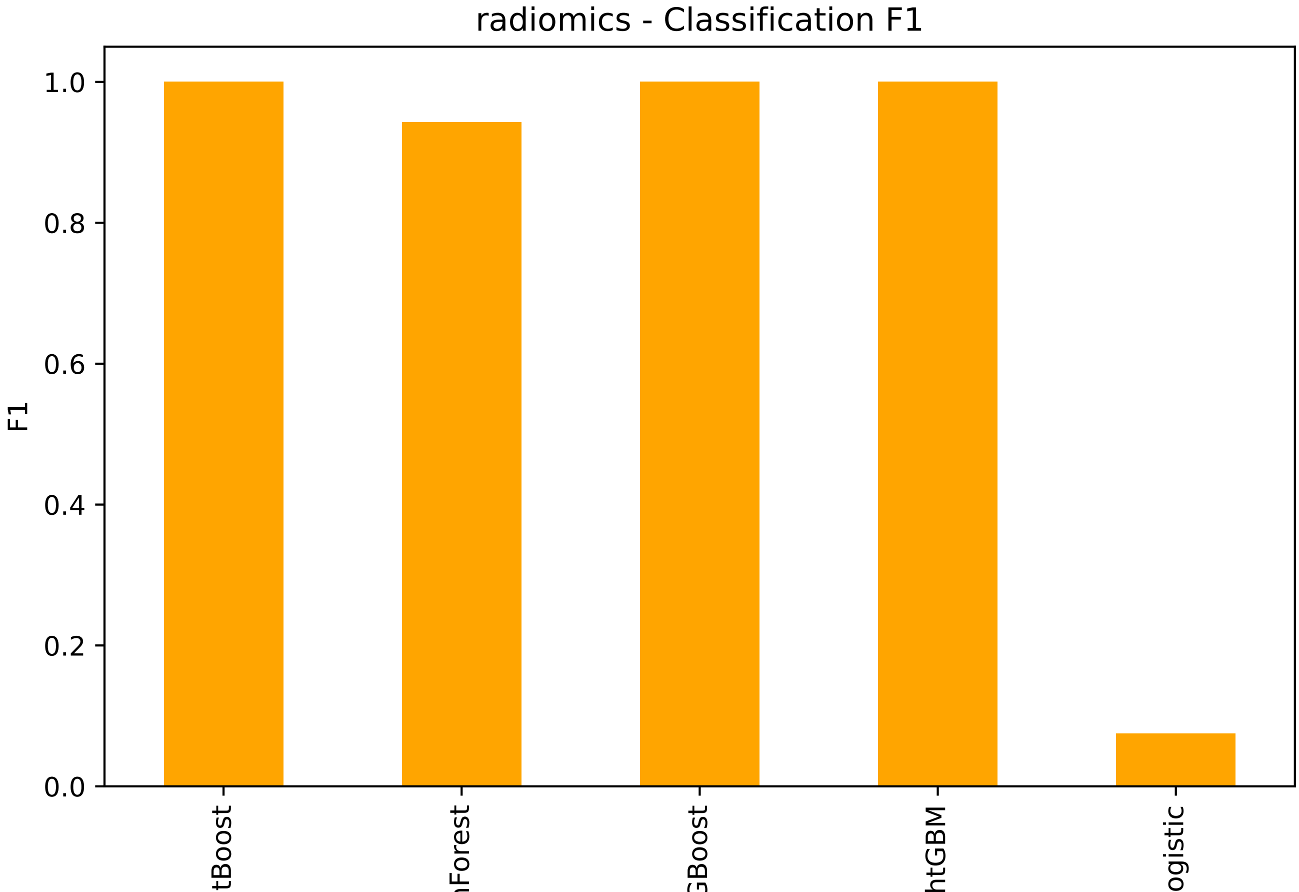

radiomics - Regression R2
R2
0
−10
−20
−30
−40
tBoost
nForest
GBoost
htGBM
Linear

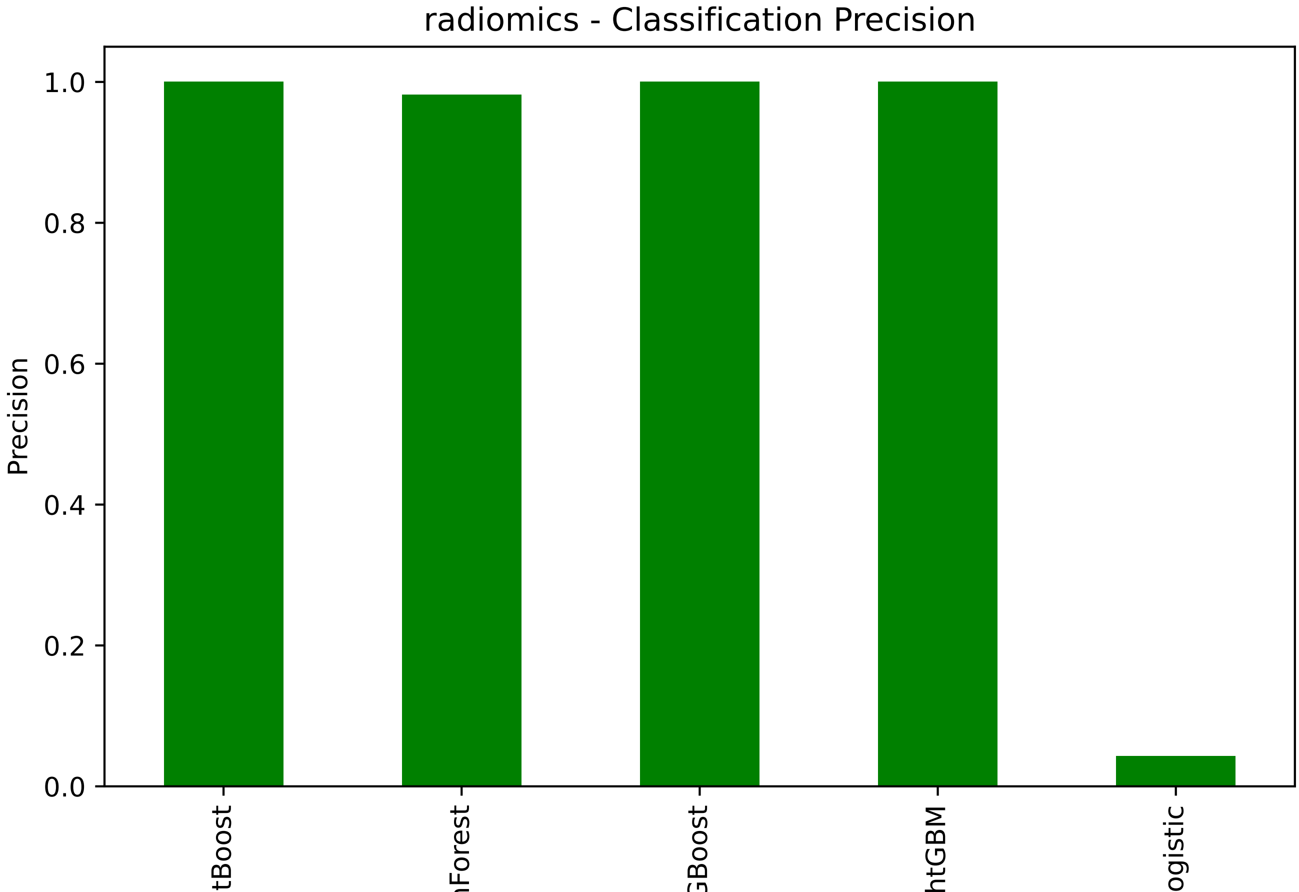

radiomics+clinical - Regression MAE
MAE
0.175
0.150
0.125
0.100
0.075
0.050
0.025
0.000
tBoost
nForest
GBoost
htGBM
Linear

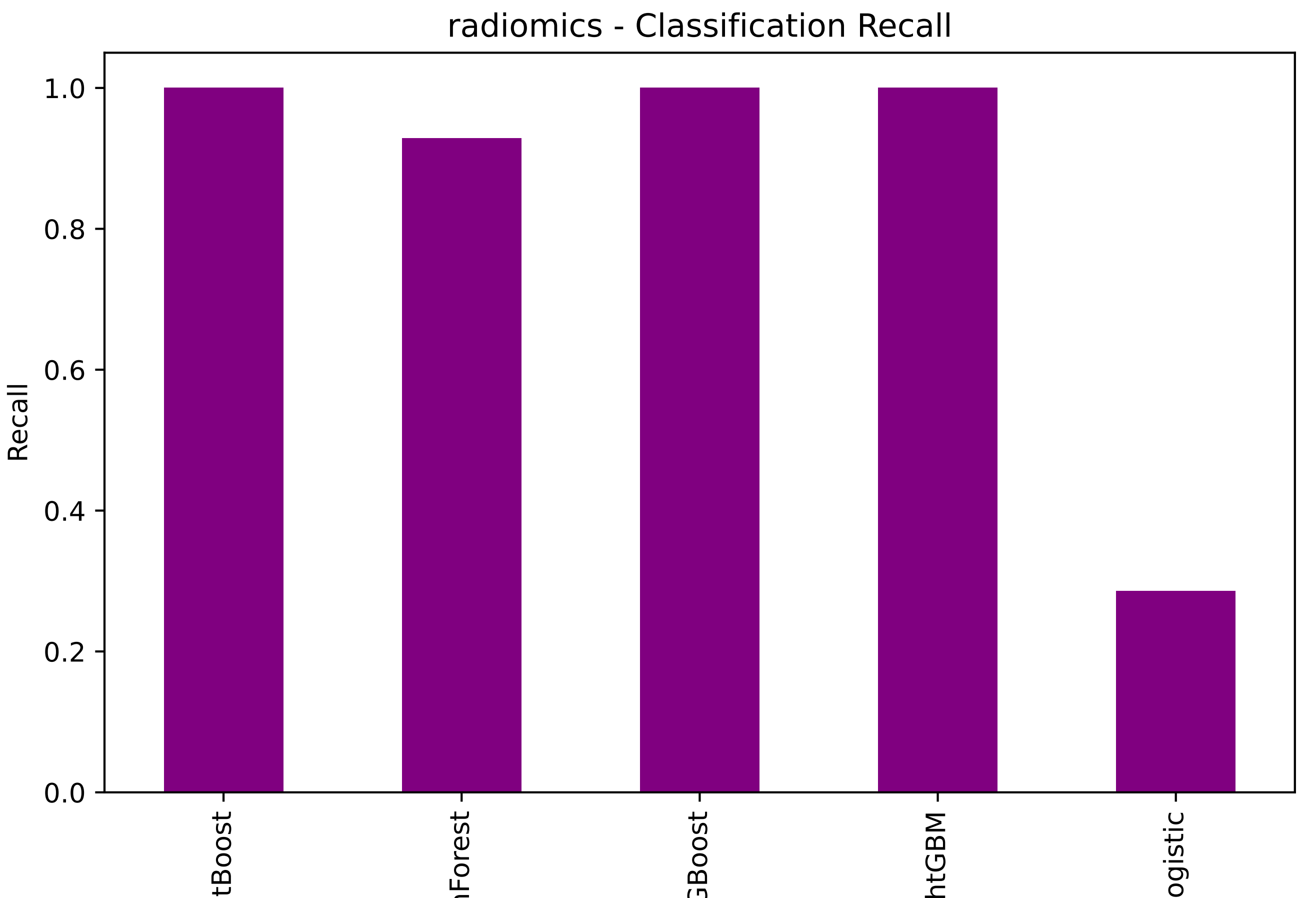

radiomics+clinical - Regression R2
R2
1.0
0.8
0.6
0.4
0.2
0.0
tBoost
Forest
GBoost
htGBM
Linear



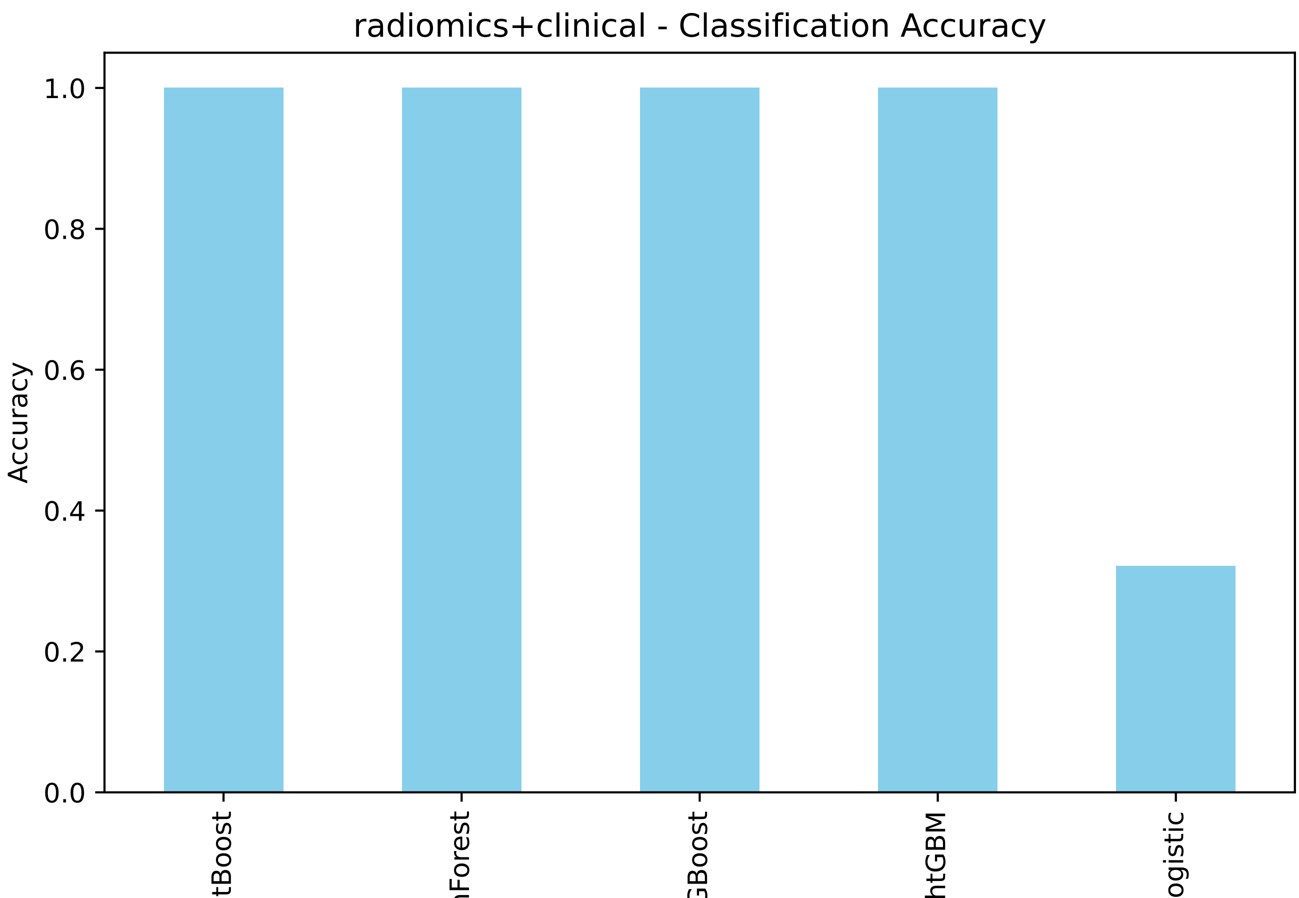
radiomics - Classification Accuracy
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
tBoost
nForest
GBoost
htGBM
ogistic

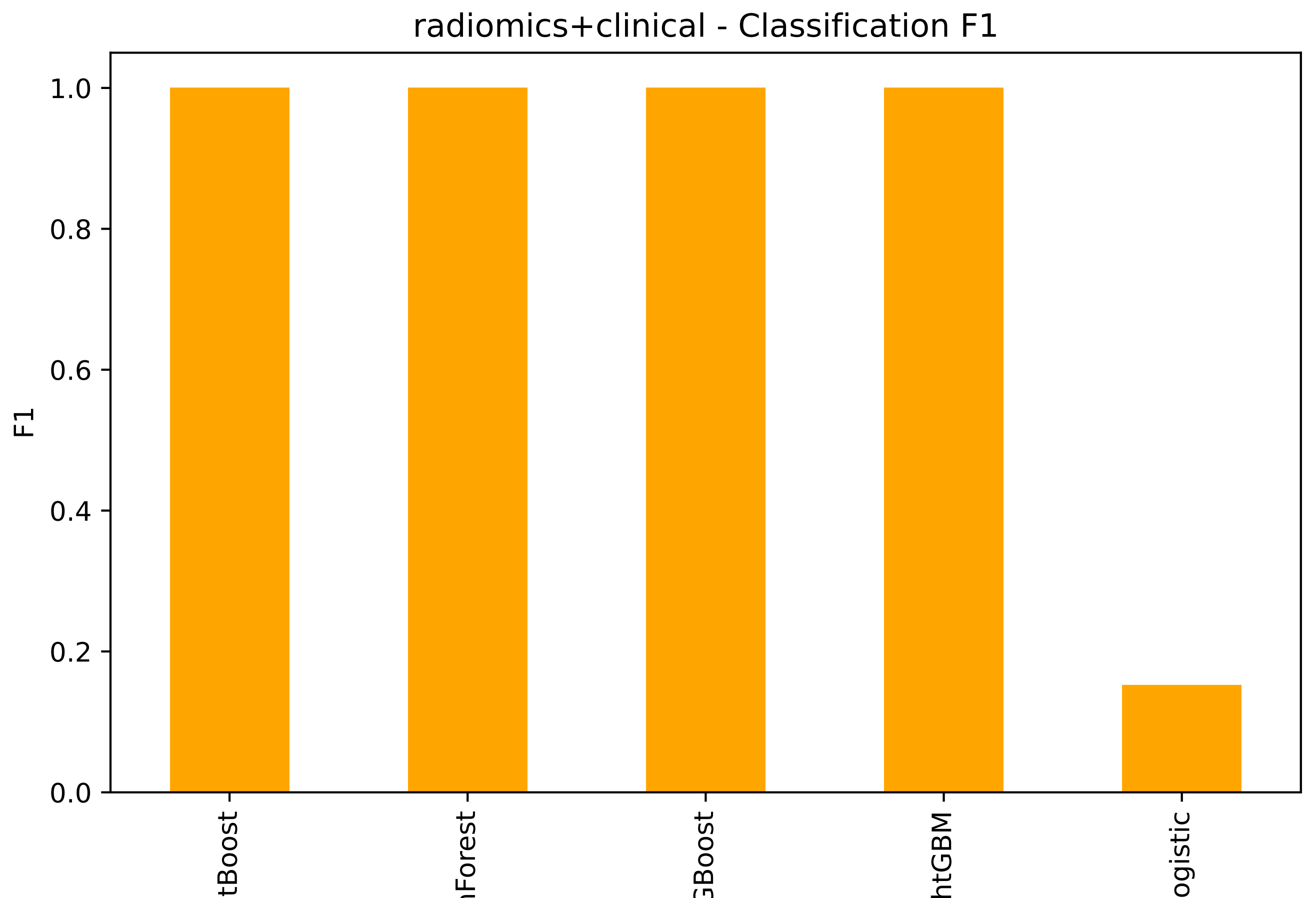

radiomics - Classification F1
F1
1.0
0.8
0.6
0.4
0.2
0.0

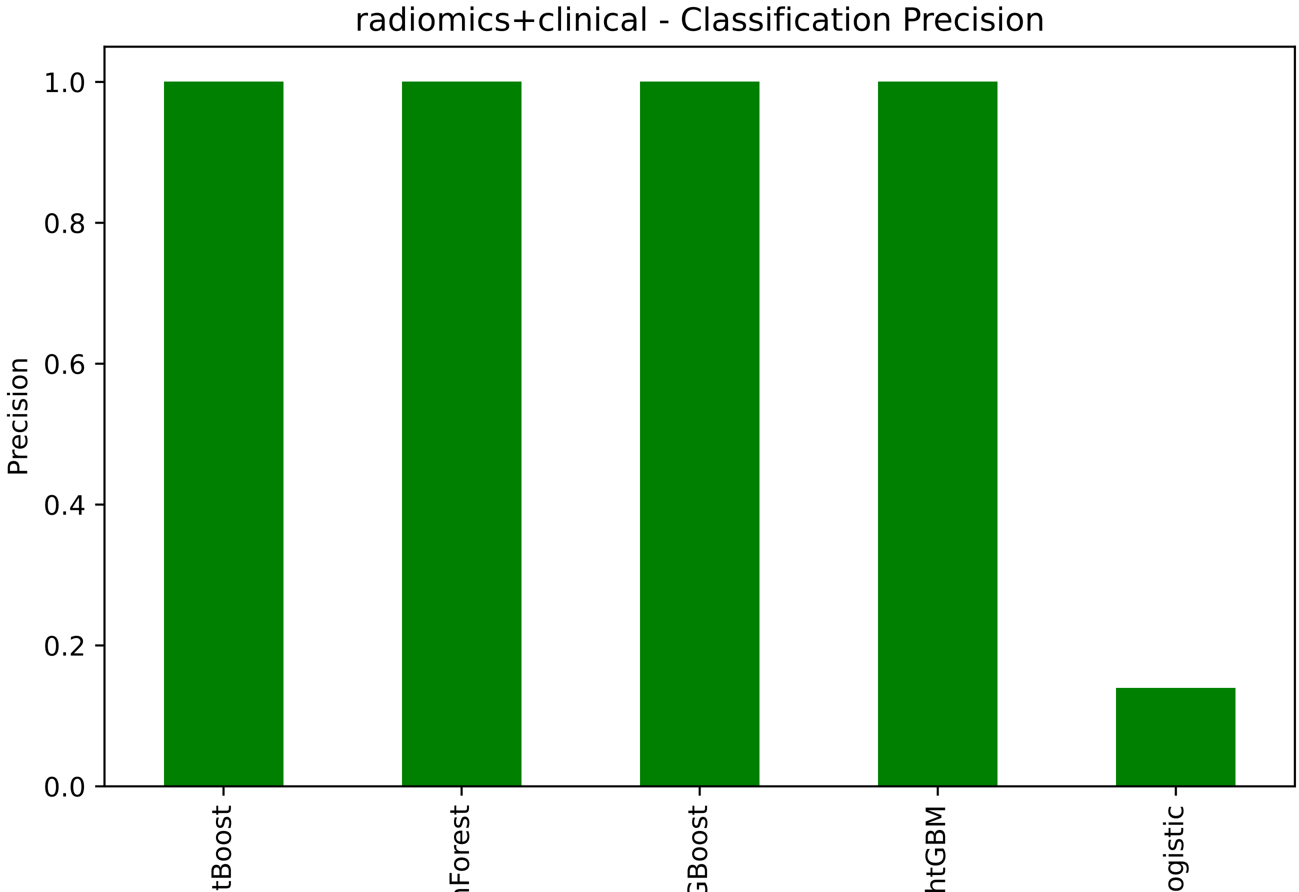

radiomics - Classification Precision
Precision
1.0
0.8
0.6
0.4
0.2
0.0
tBoost
nForest
GBoost
htGBM
ogistic

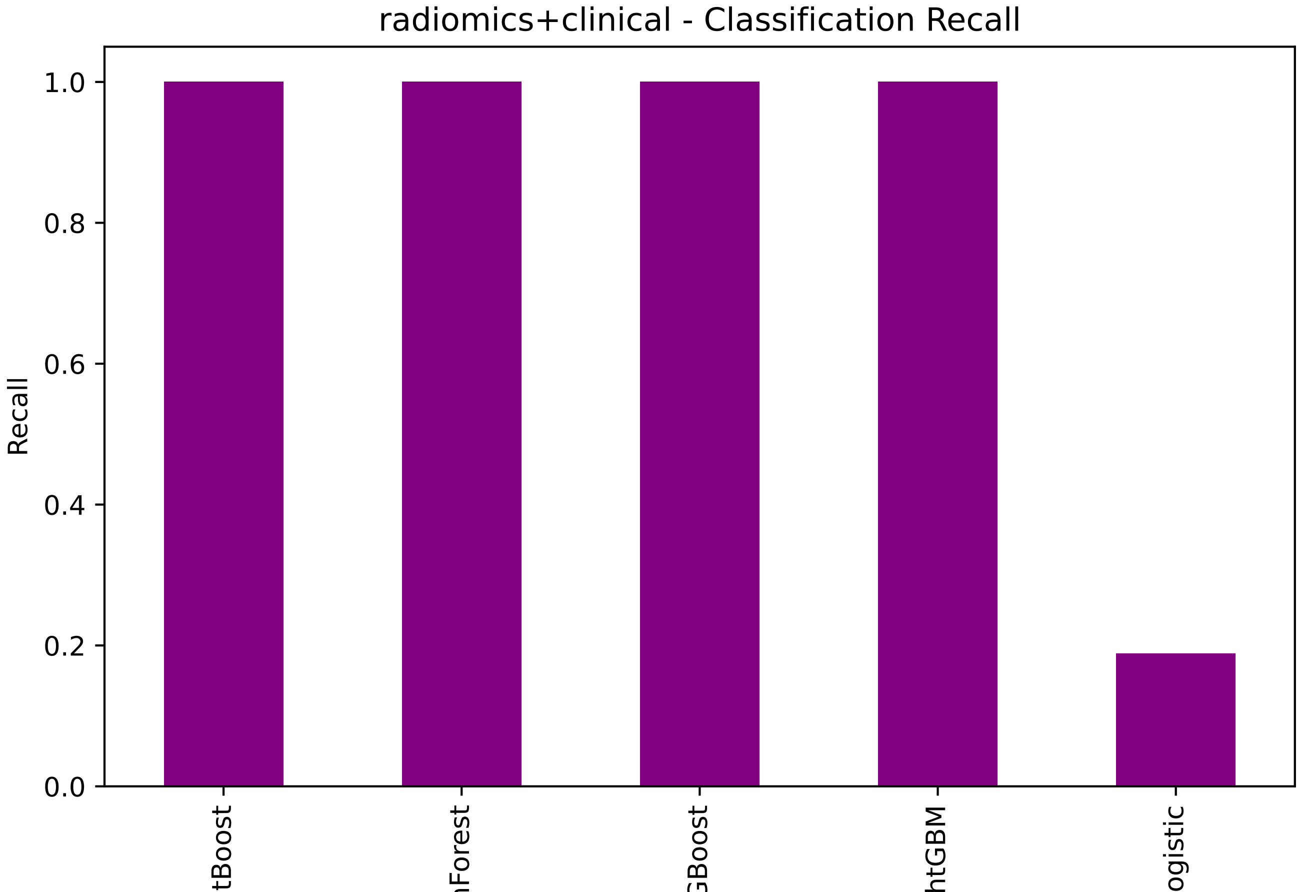


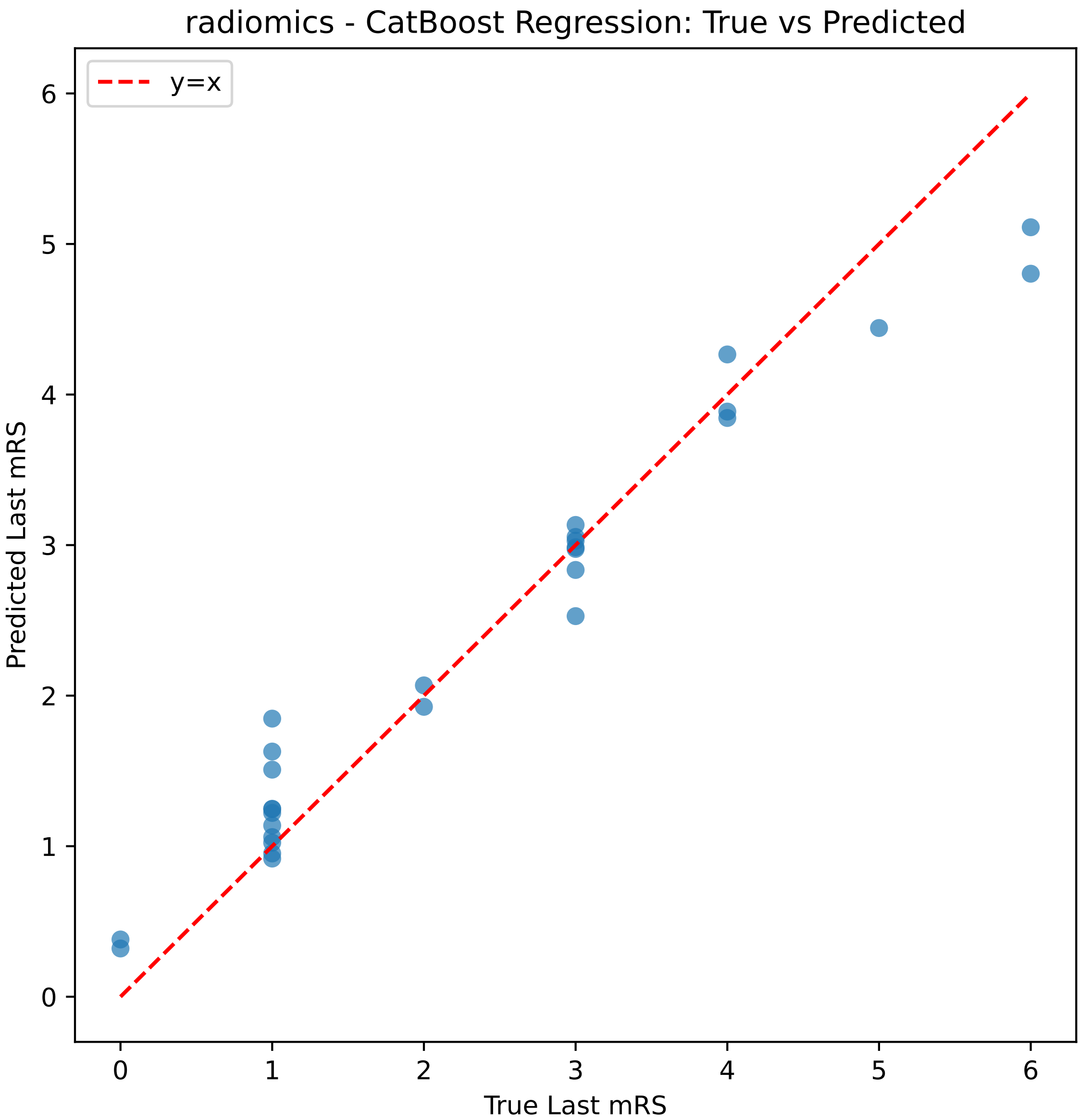
radiomics - Classification Recall
Recall
1.0
0.8
0.6
0.4
0.2
0.0
tBoost
nForest
GBoost
htGBM
ogistic

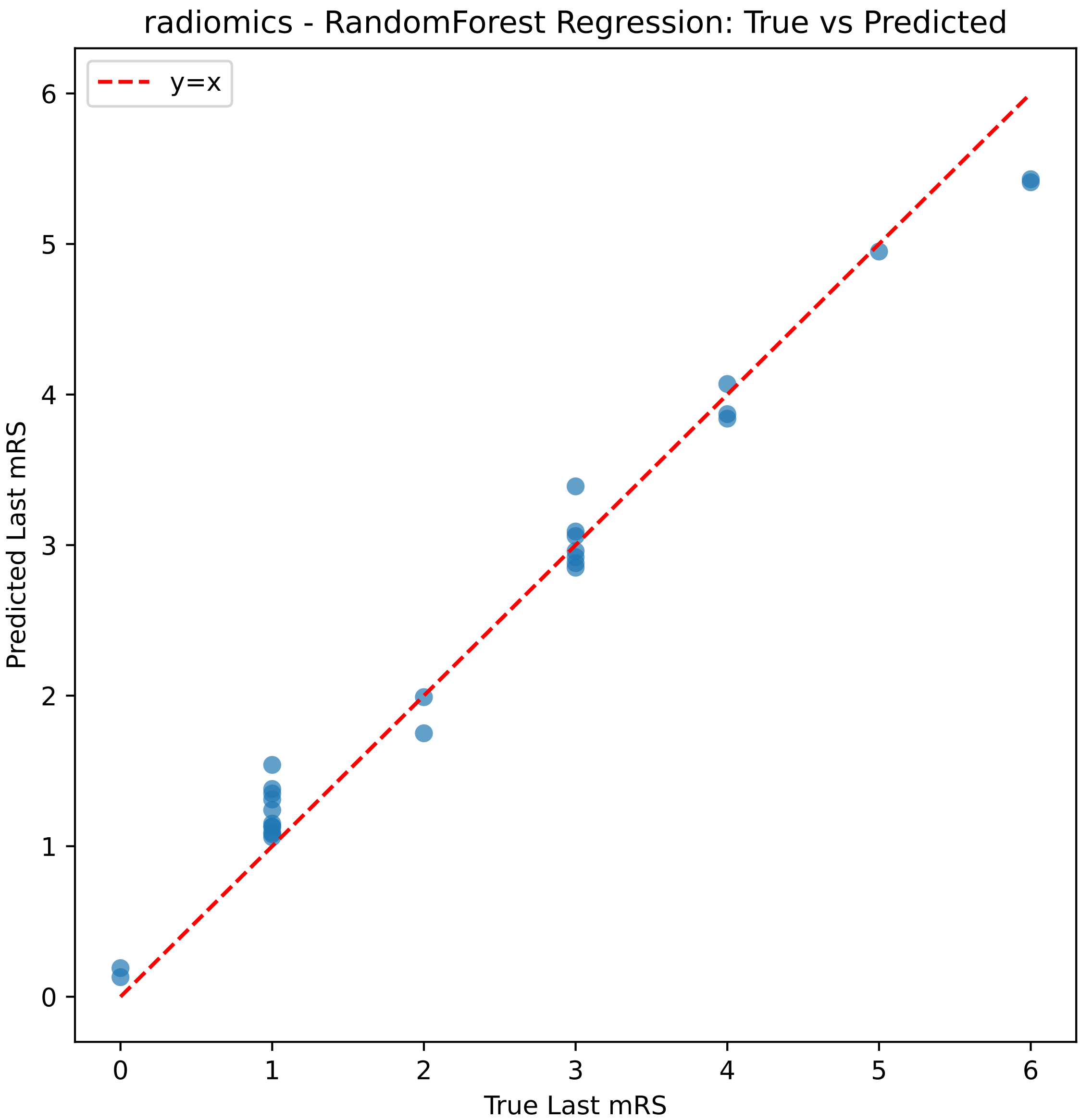

radiomics+clinical - Classification Accuracy
Accuracy
1.0
0.8
0.6
0.4
0.2
0.0
tBoost
nForest
GBoost
htGBM
ogistic

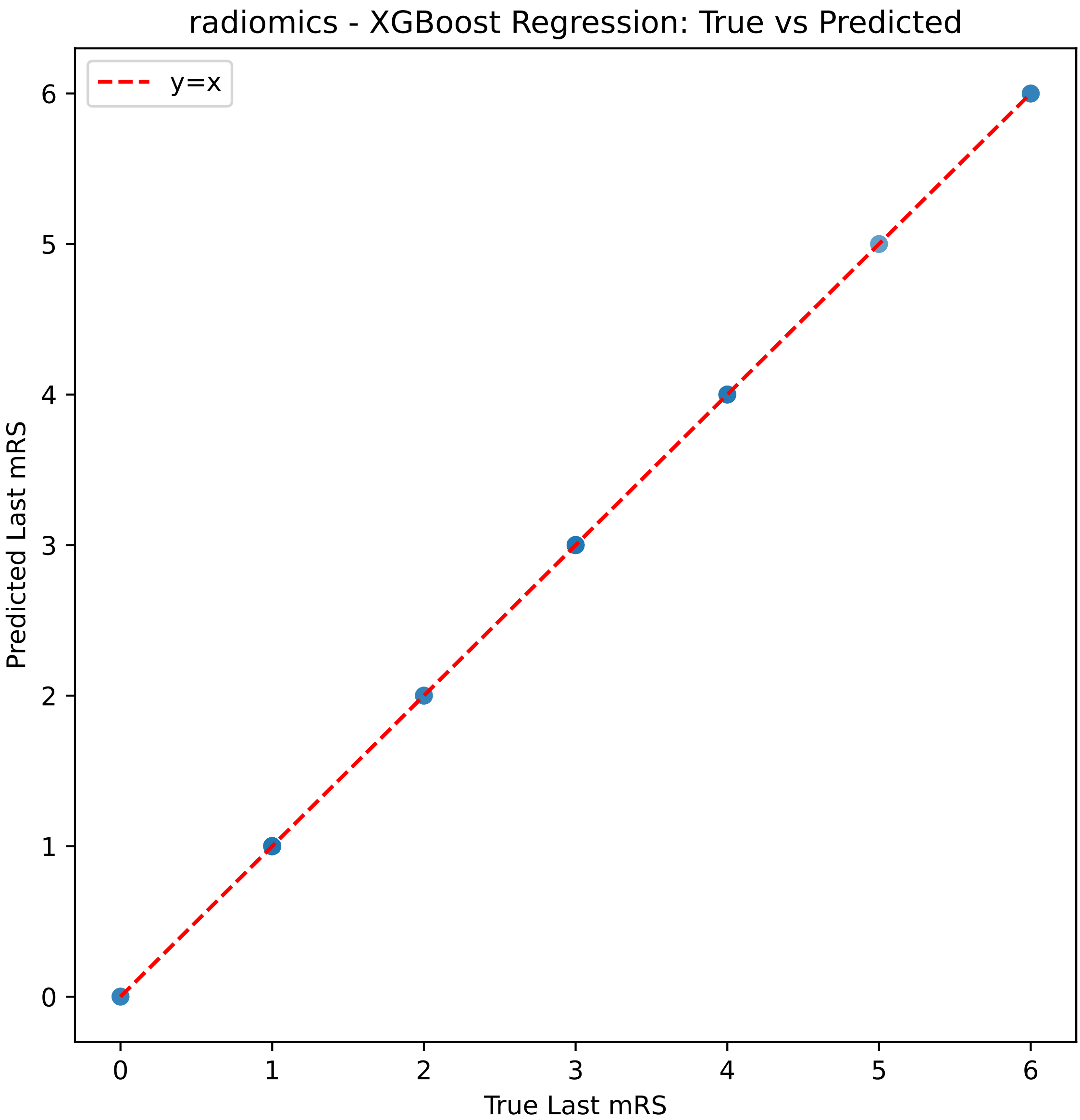

radiomics+clinical - Classification F1
F1
1.0
0.8
0.6
0.4
0.2
0.0
tBoost
nForest
GBoost
htGBM
ogistic


radiomics+clinical - Classification Precision
Precision
1.0
0.8
0.6
0.4
0.2
0.0
tBoost
nForest
GBoost
htGBM
ogistic

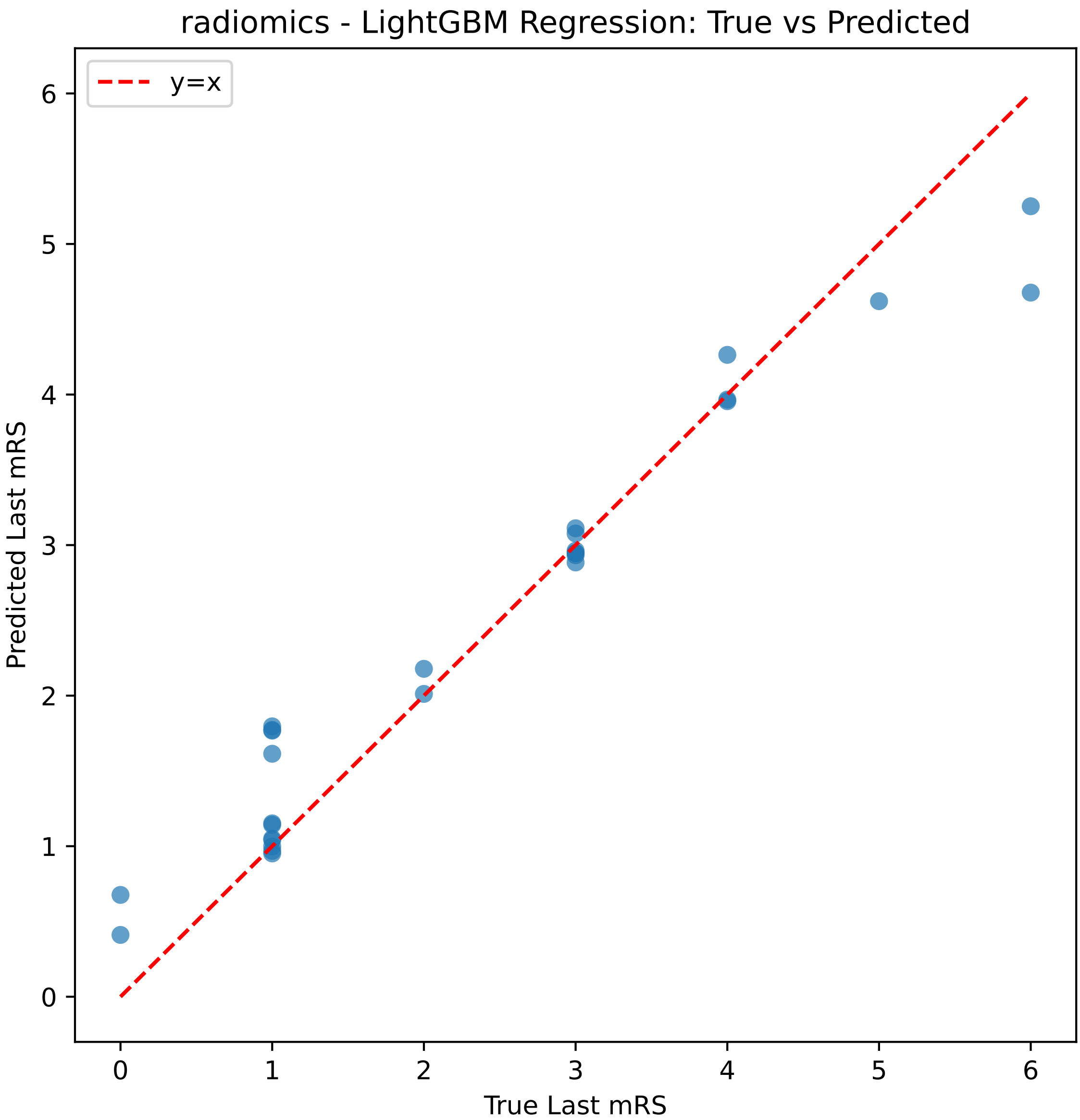

radiomics+clinical - Classification Recall
Recall
1.0
0.8
0.6
0.4
0.2
0.0

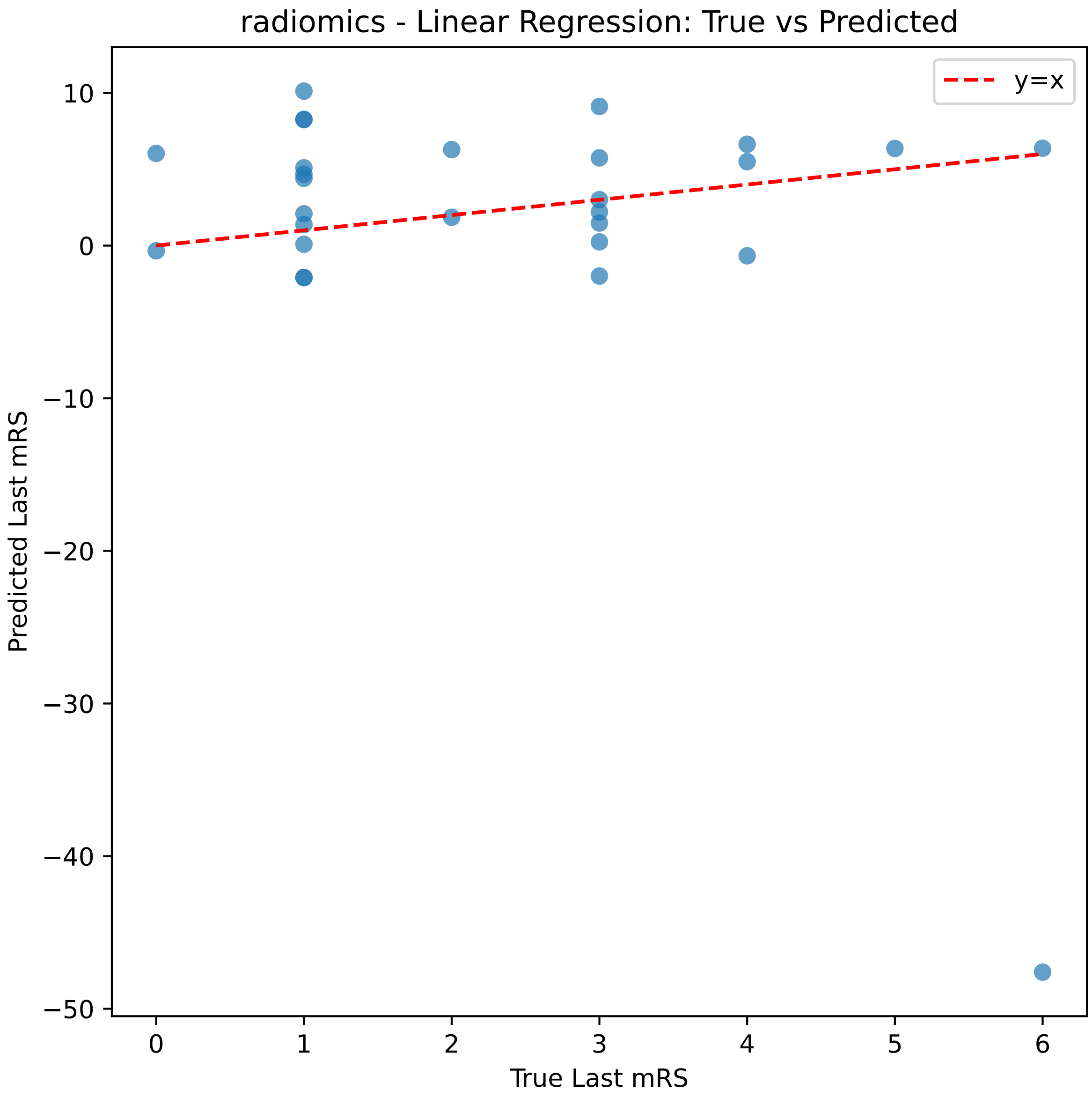

radiomics - CatBoost Regression: True vs Predicted
y=x
Predicted Last mRS
True Last mRS
0
1
2
3
4
5
6
0
1
2
3
4
5
6

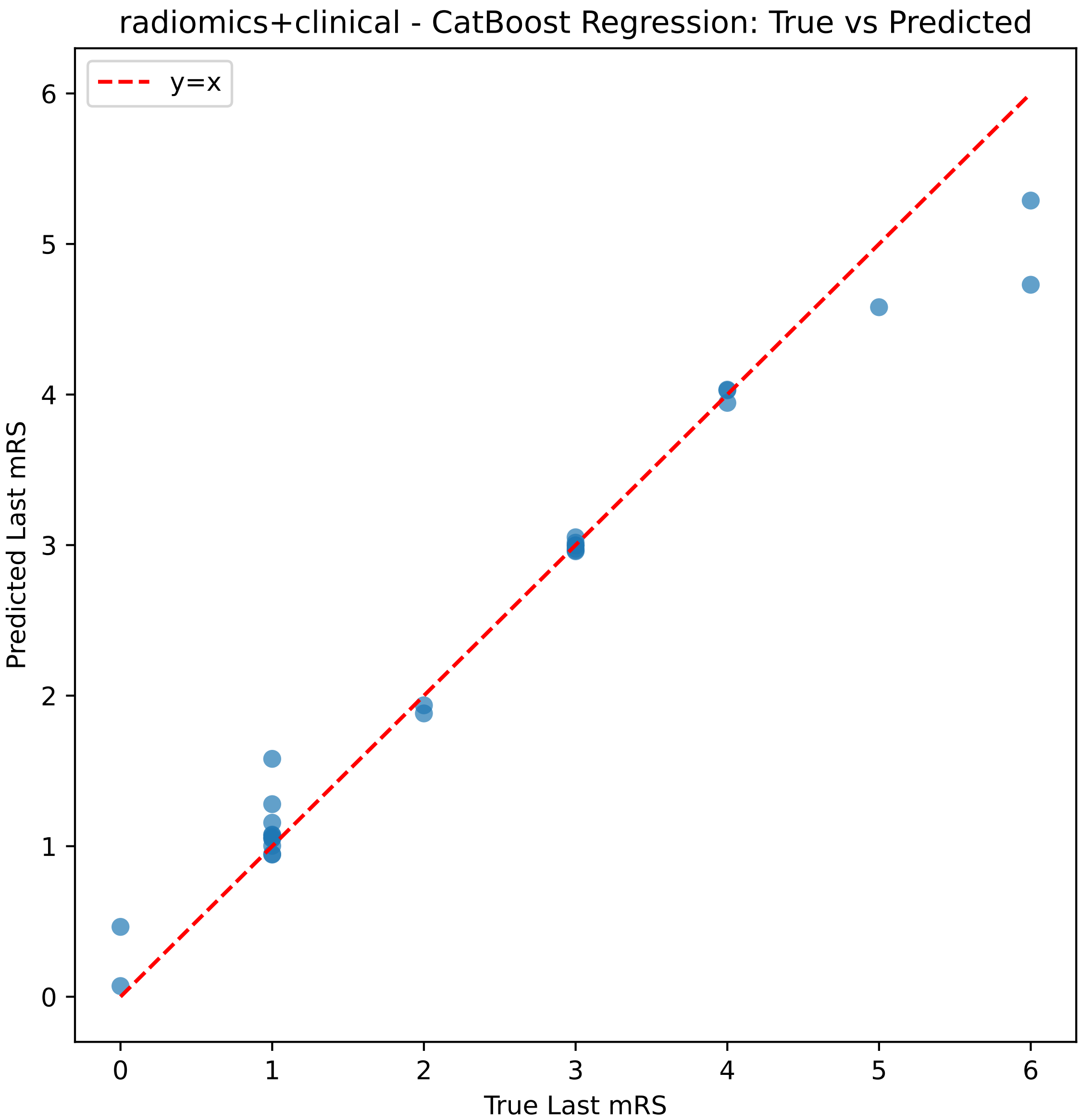

radiomics - RandomForest Regression: True vs Predicted
y=x
Predicted Last mRS
True Last mRS
0
1
2
3
4
5
6



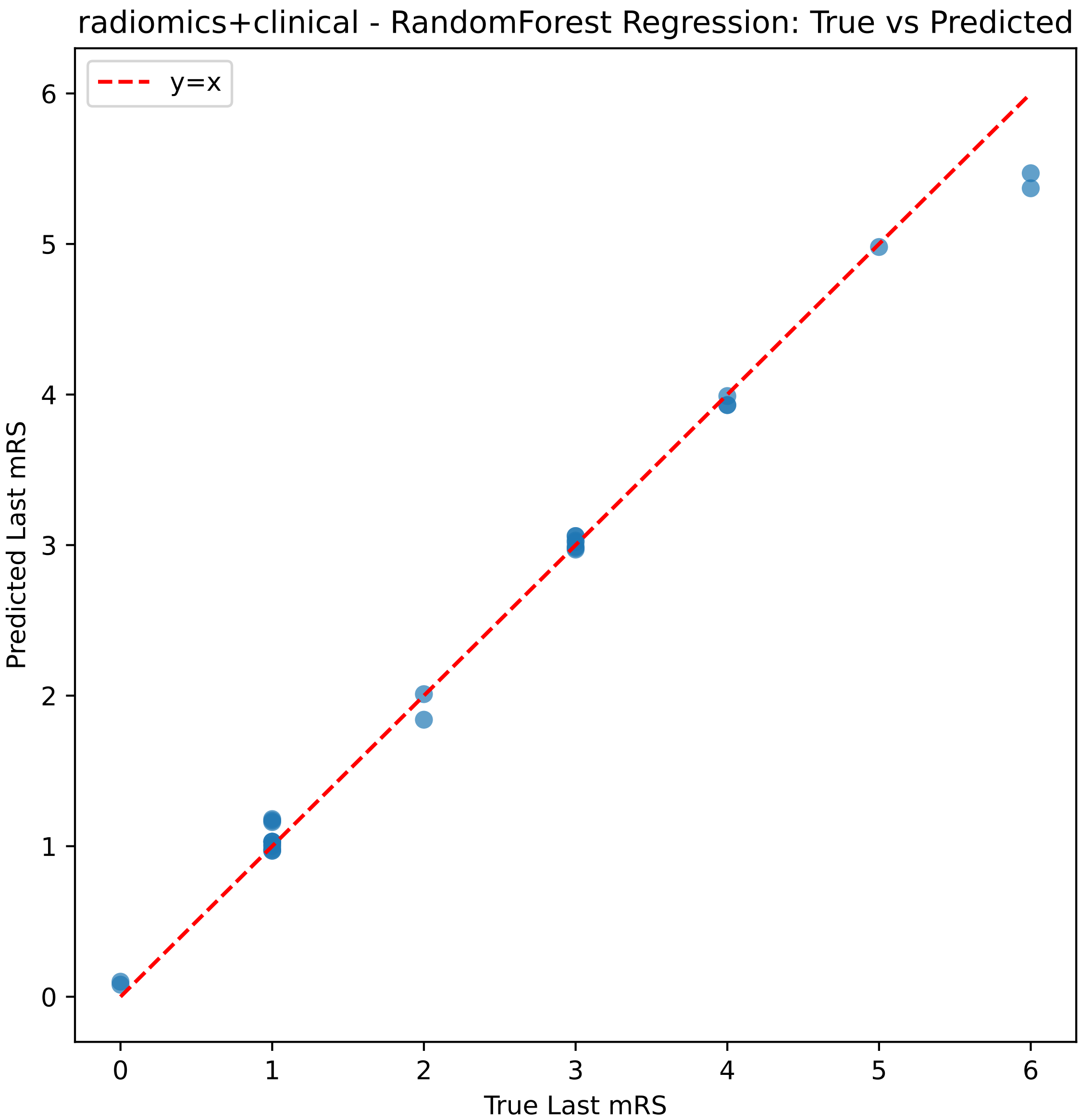
radiomics - XGBoost Regression: True vs Predicted
y=x
Predicted Last mRS
True Last mRS
0
1
2
3
4
5
6

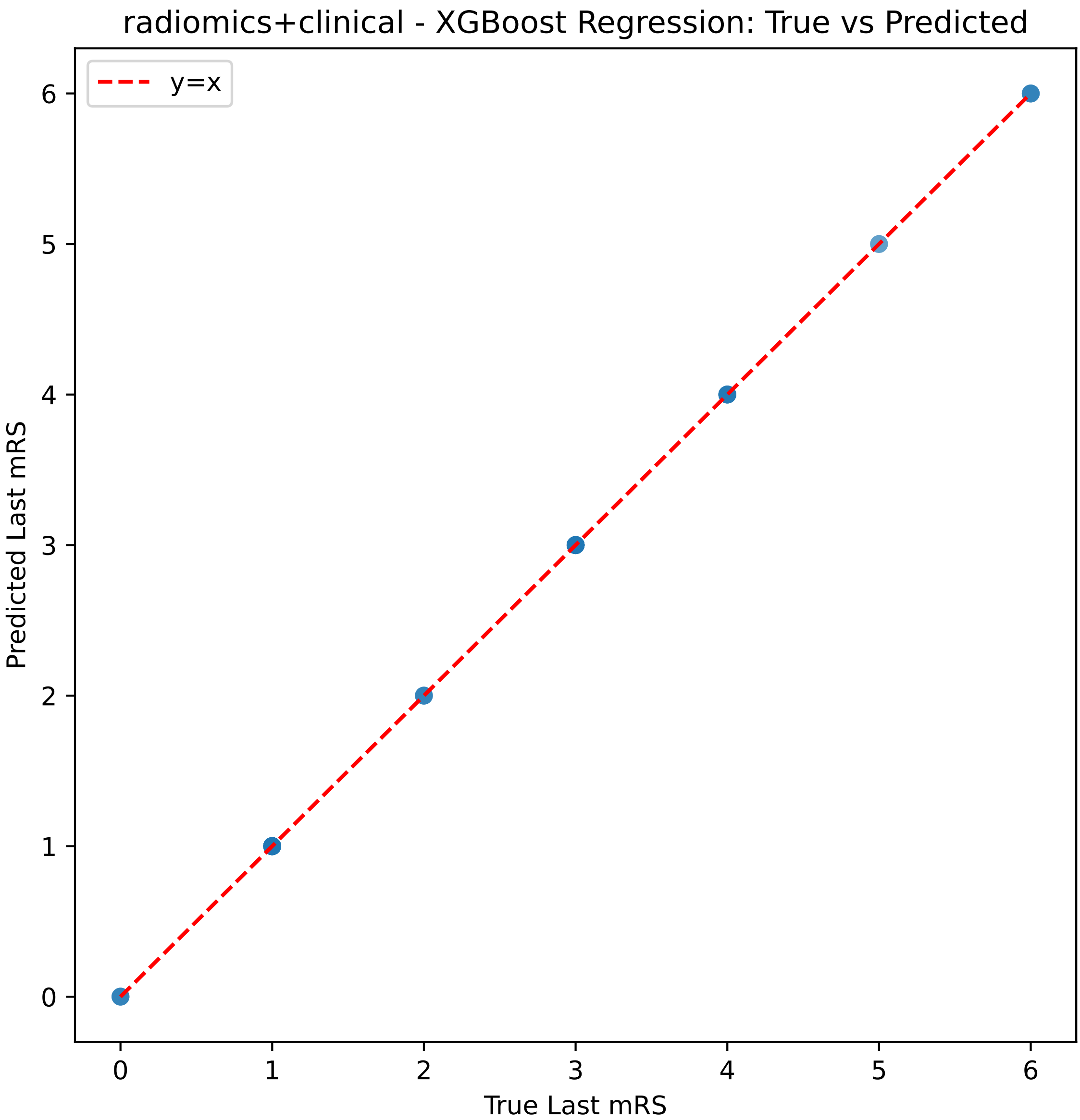

radiomics - LightGBM Regression: True vs Predicted
y=x
Predicted Last mRS
True Last mRS
0
1
2
3
4
5
6

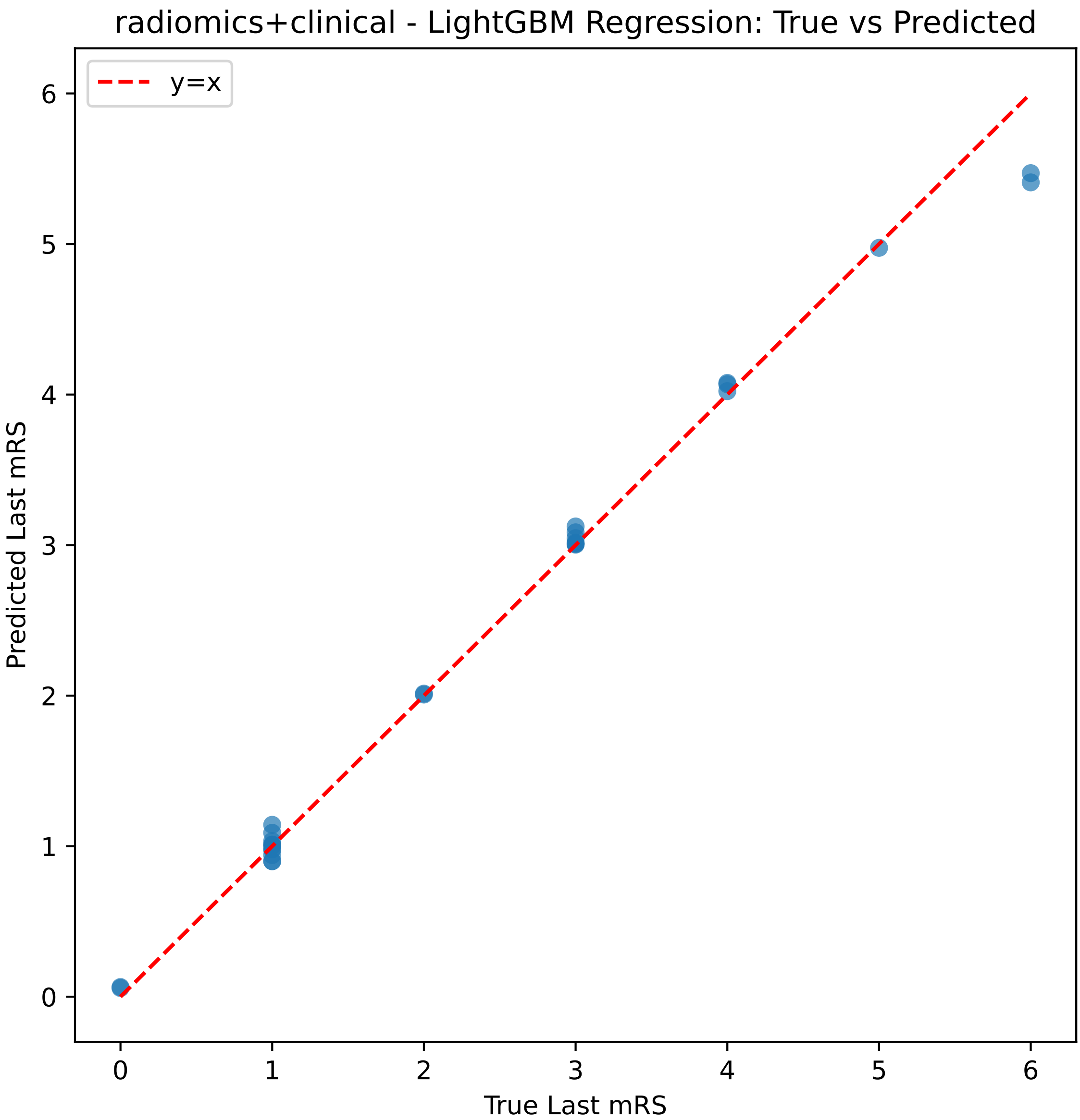

radiomics - Linear Regression: True vs Predicted
y=x
Predicted Last mRS
True Last mRS
10
0
−10
−20
−30
−40
−50
0
1
2
3
4
5
6

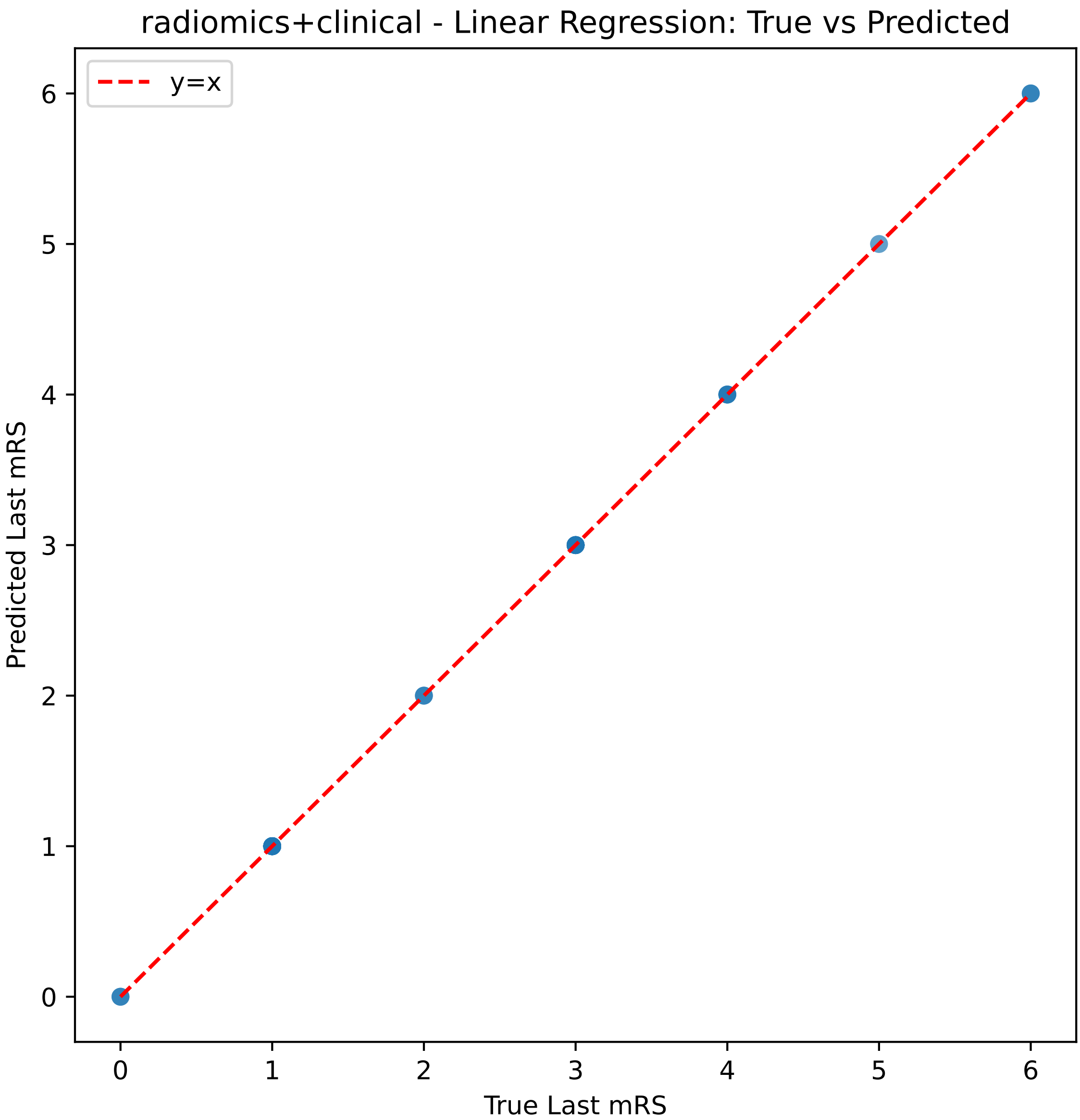


radiomics+clinical - CatBoost Regression: True vs Predicted
y=x
Predicted Last mRS
True Last mRS
0
1
2
3
4
5
6
0
1
2
3
4
5
6

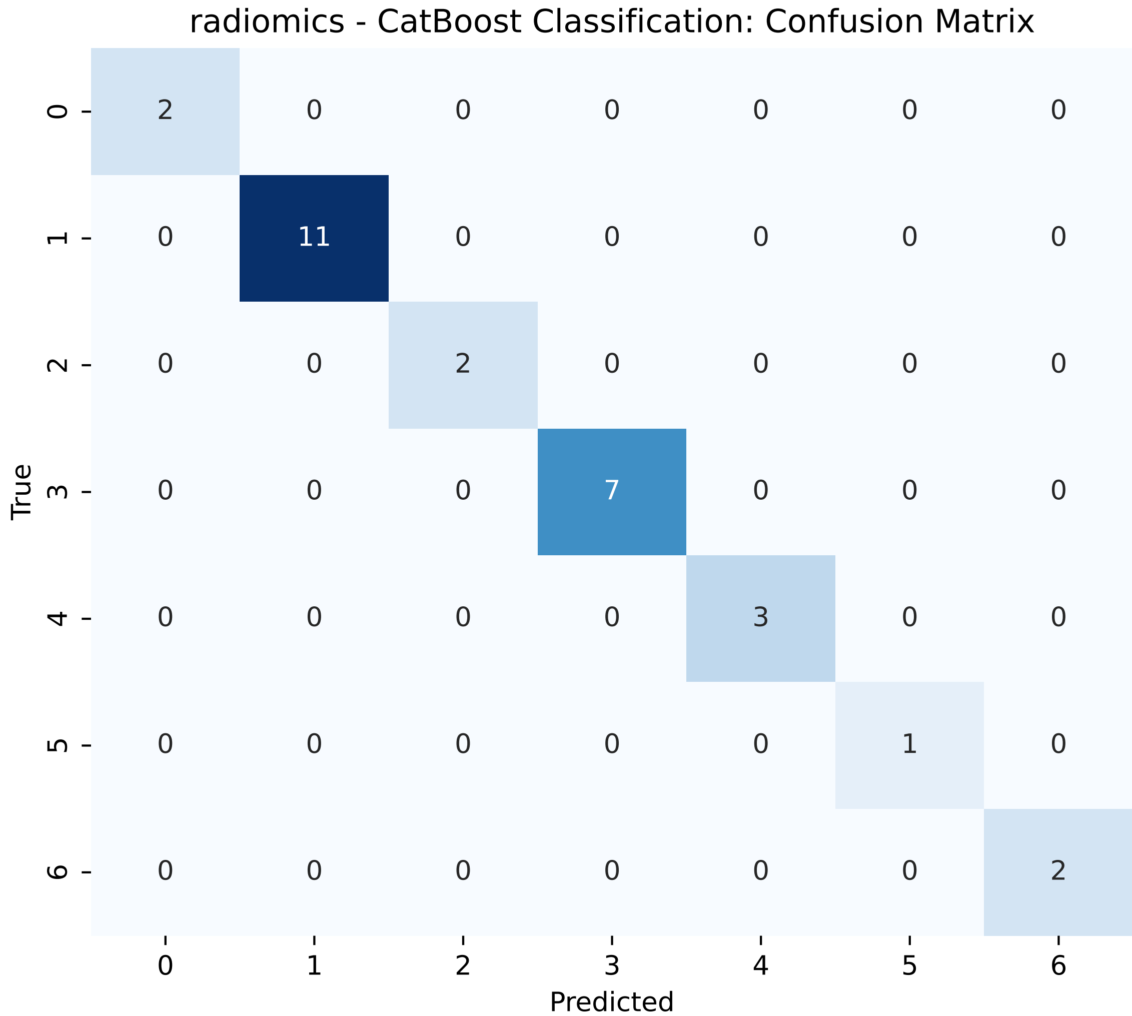

radiomics+clinical - RandomForest Regression: True vs Predicted
y=x
Predicted Last mRS
True Last mRS
0
1
2
3
4
5
6
0
1
2
3
4
5
6


radiomics+clinical - XGBoost Regression: True vs Predicted
y=x
Predicted Last mRS
True Last mRS
0
1
2
3
4
5
6

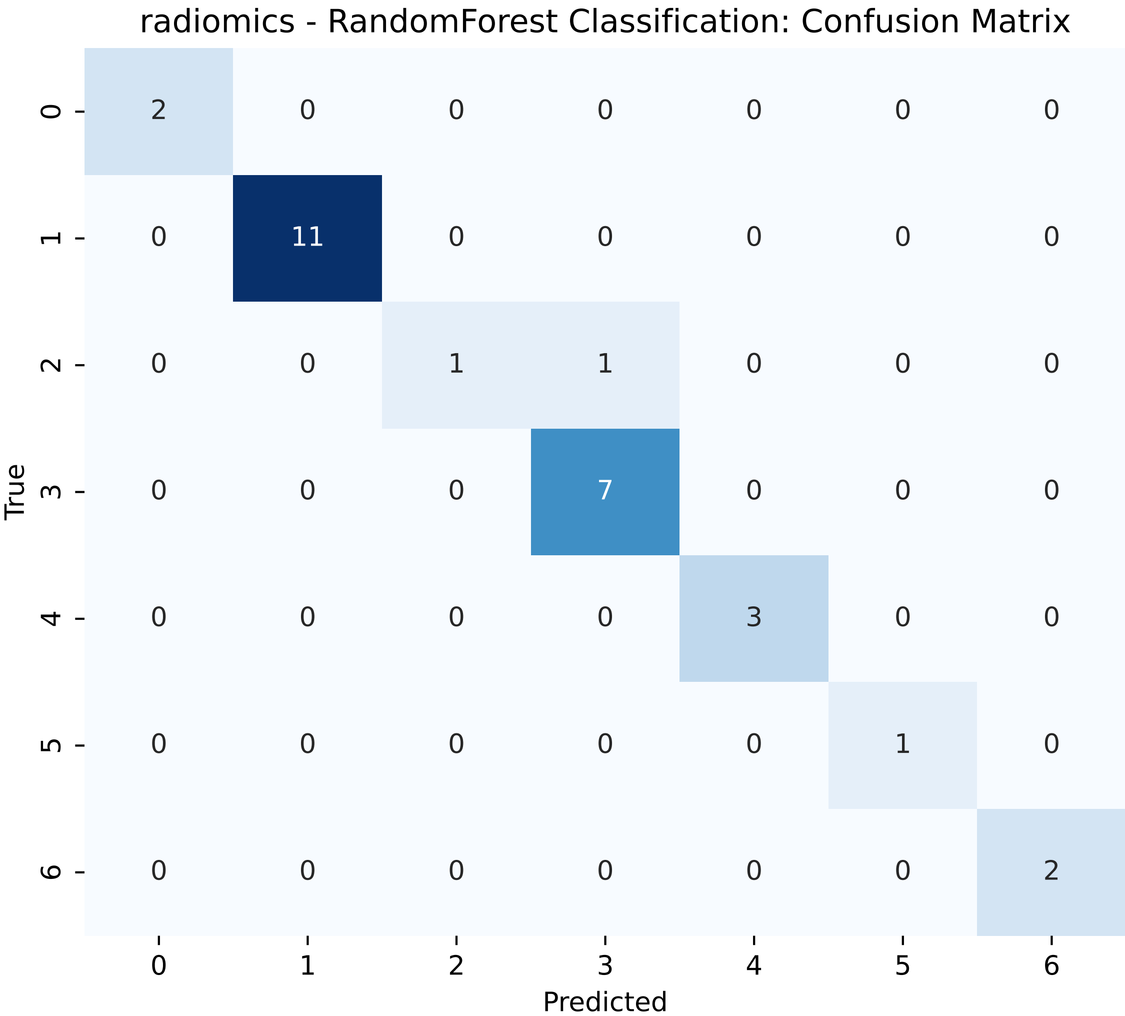

radiomics+clinical - LightGBM Regression: True vs Predicted
y=x
Predicted Last mRS
True Last mRS
0
1
2
3
4
5
6



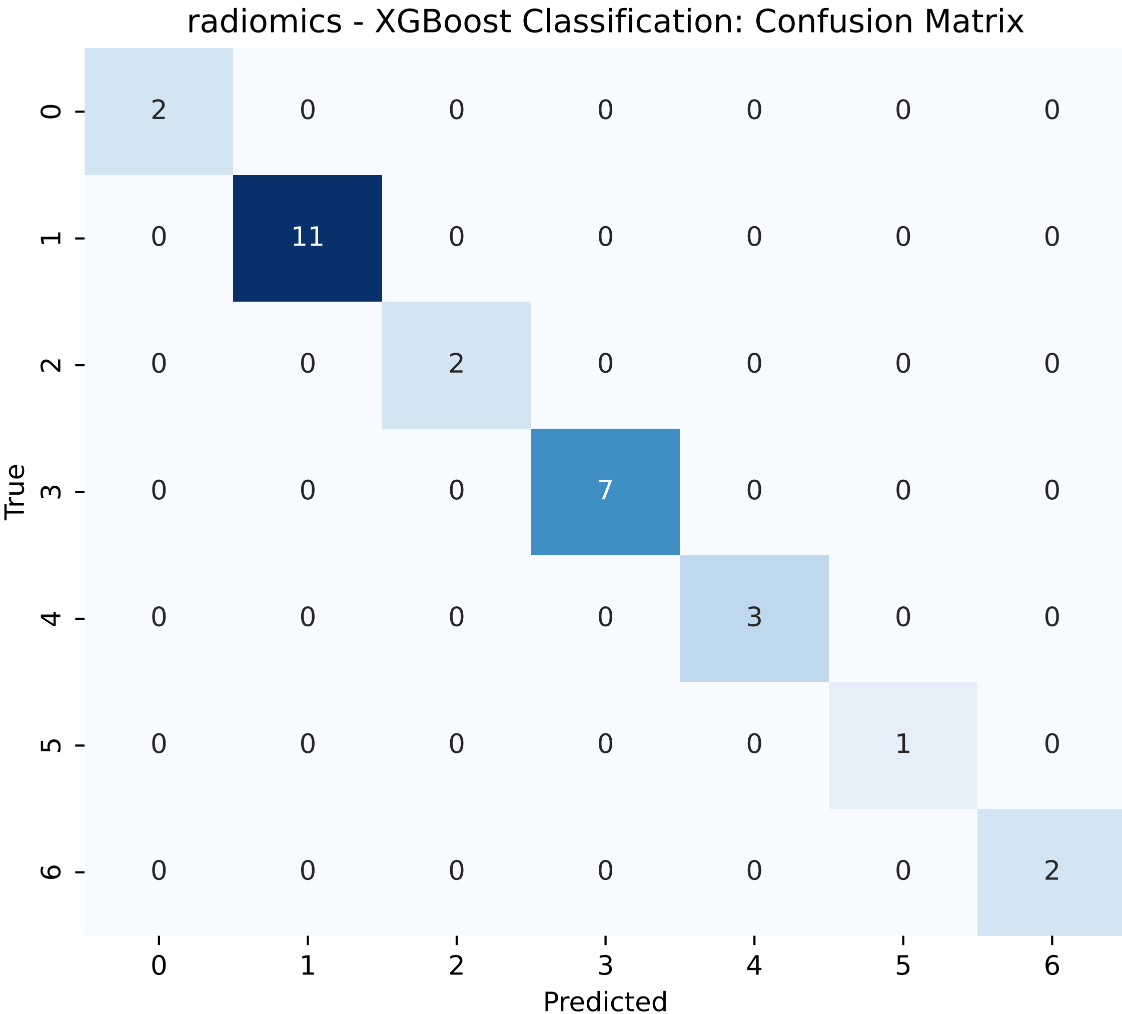
radiomics+clinical - Linear Regression: True vs Predicted
y=x
Predicted Last mRS
True Last mRS
0
1
2
3
4
5
6

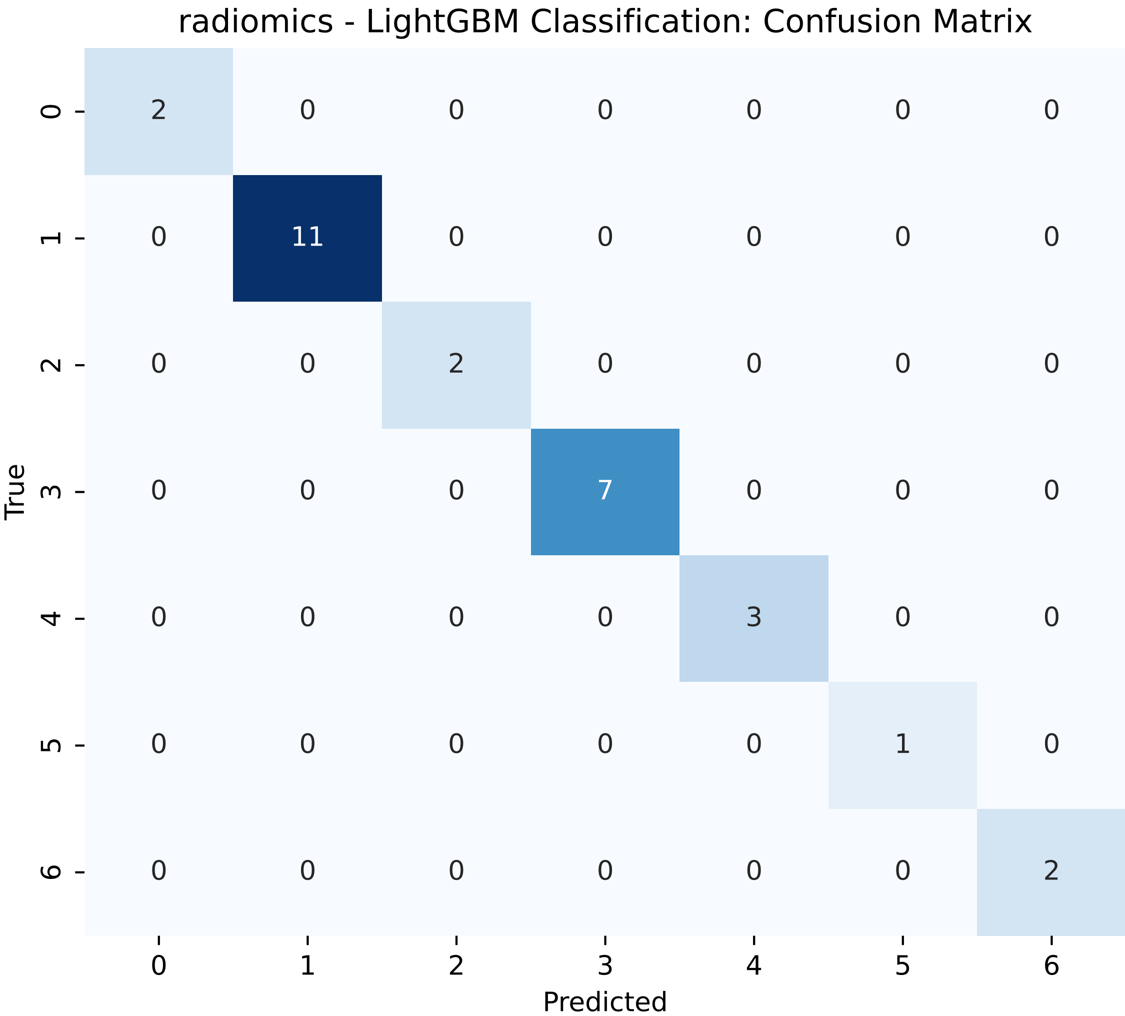


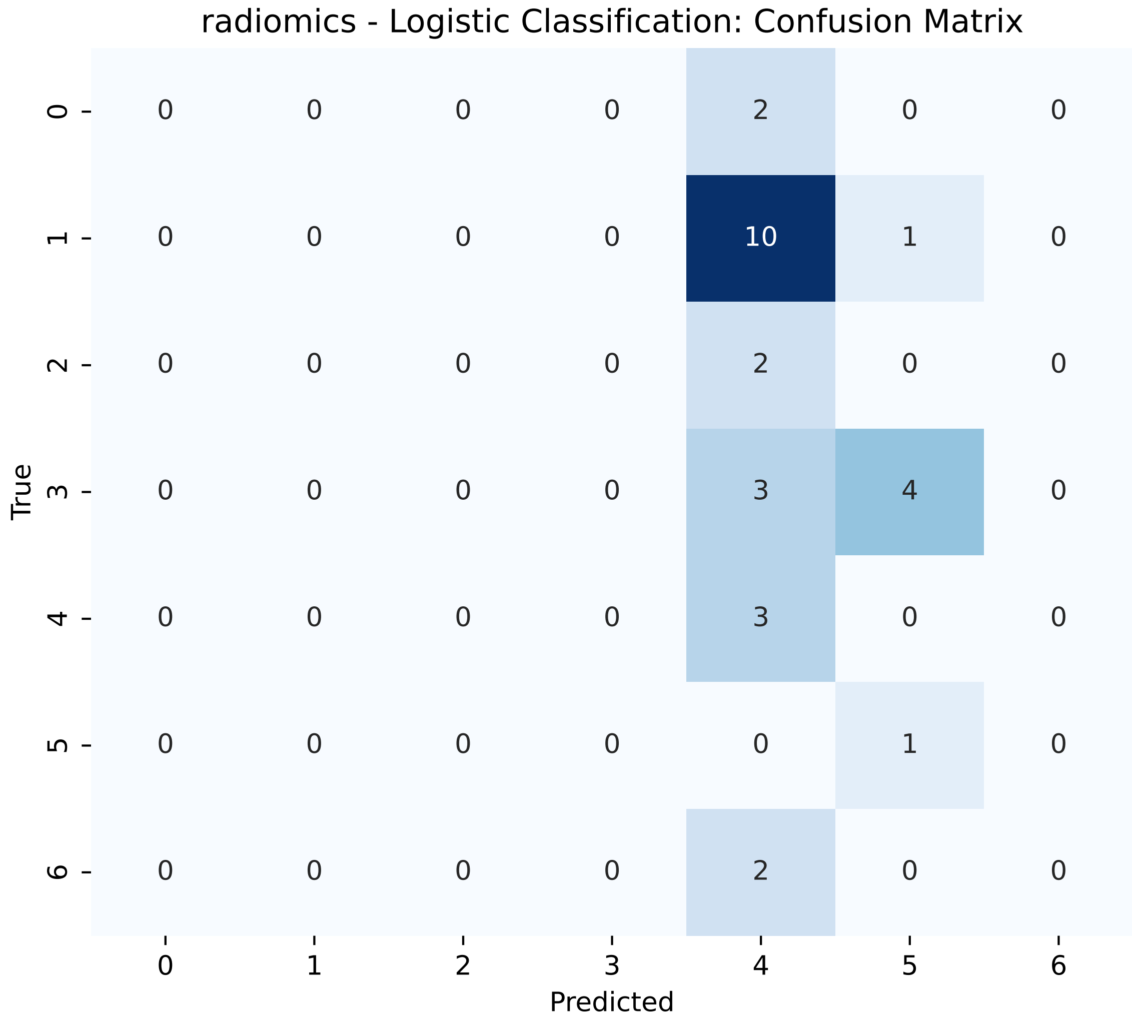
radiomics - CatBoost Classification: Confusion Matrix
True
Predicted
0
1
2
3
4
5
6
2
11
7
3
1

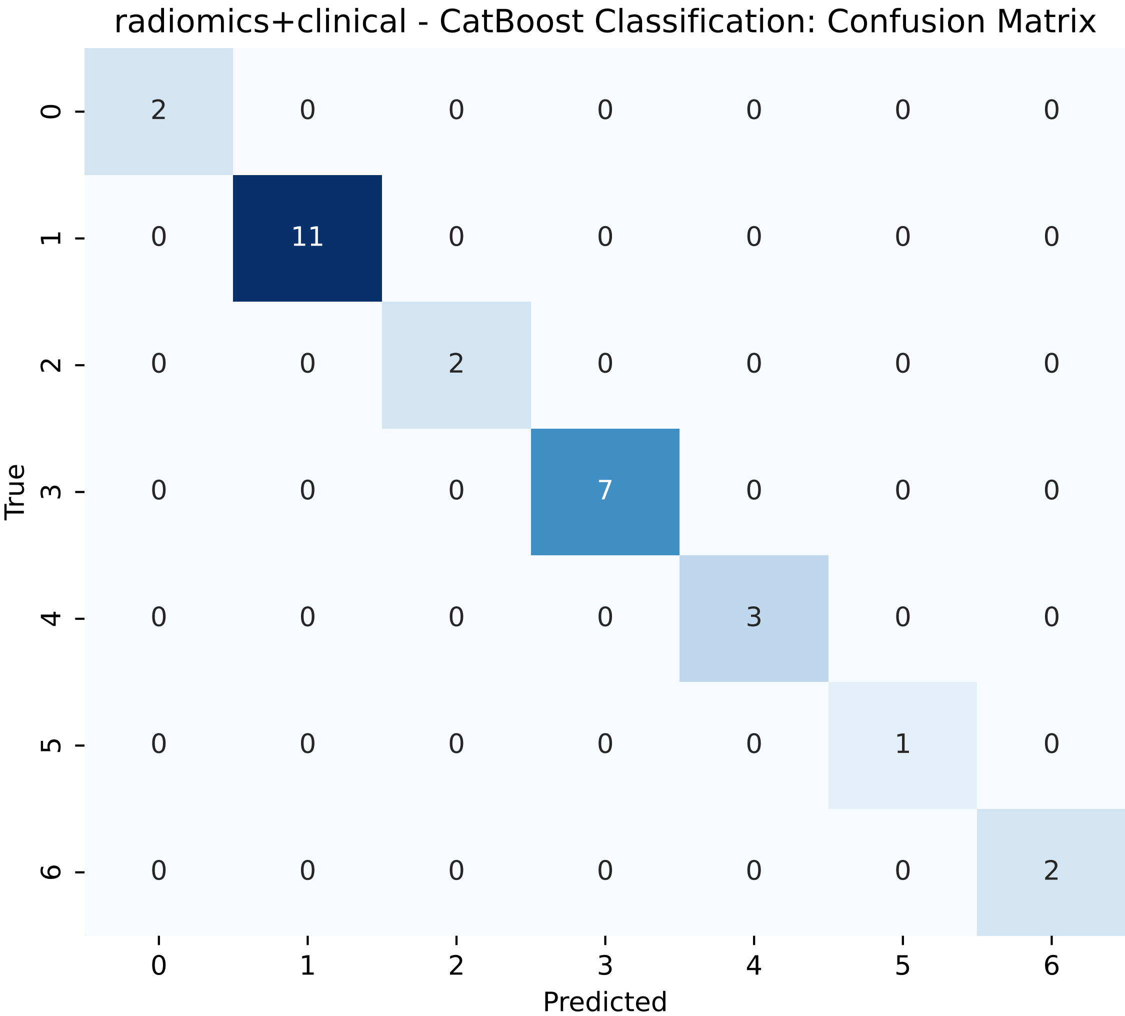


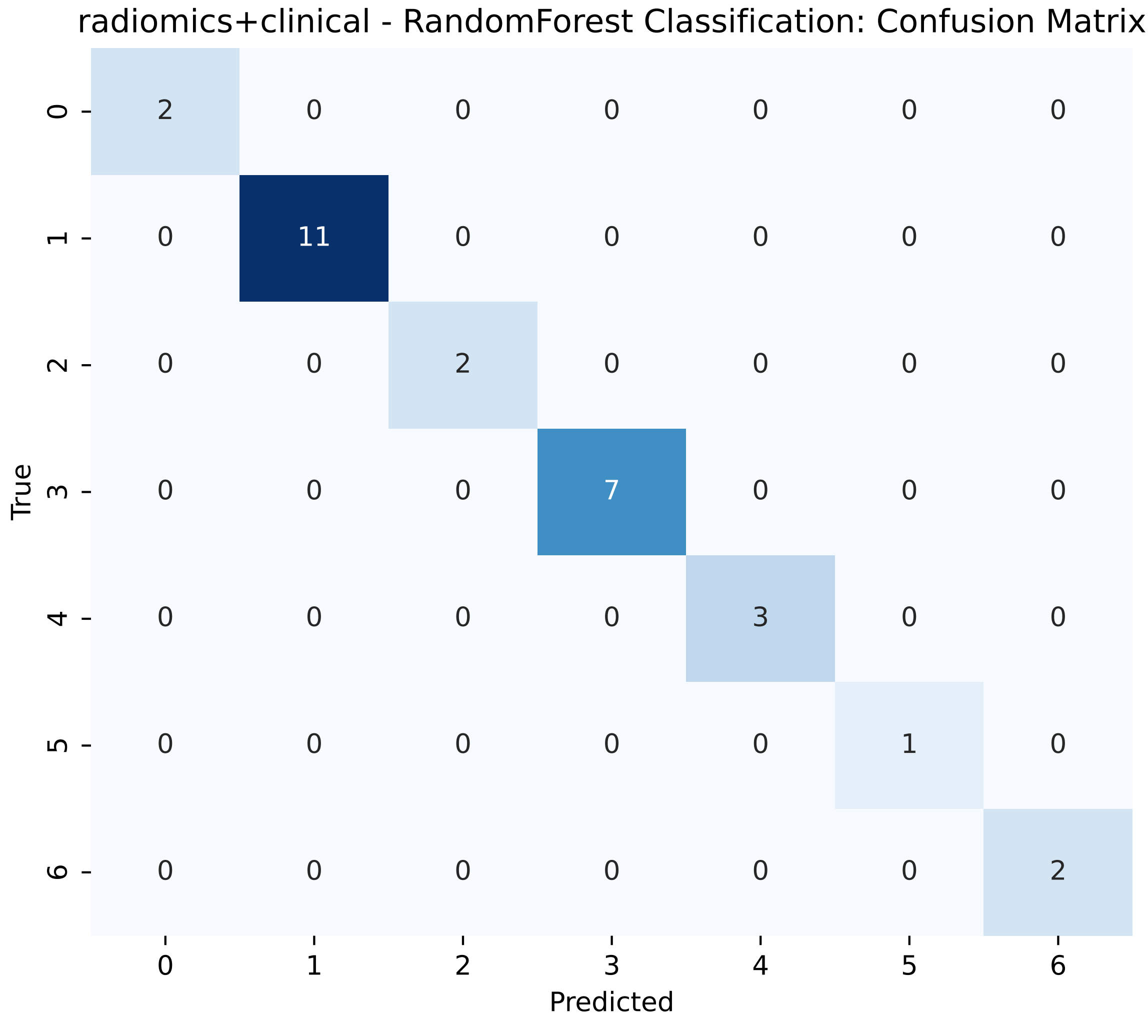
radiomics - RandomForest Classification: Confusion Matrix
True
Predicted
0
1
2
3
4
5
6
2 0 0 0 0 0 0
0 11 0 0 0 0 0
0 0 1 1 0 0 0
0 0 0 7 0 0 0
0 0 0 0 3 0 0
0 0 0 0 0 1 0
0 0 0 0 0 0 2


radiomics - XGBoost Classification: Confusion Matrix
True
Predicted
0
1
2
3
4
5
6
2 0 0 0 0 0 0
0 11 0 0 0 0 0
0 0 2 0 0 0 0
0 0 0 7 0 0 0
0 0 0 0 3 0 0
0 0 0 0 0 1 0
0 0 0 0 0 0 2

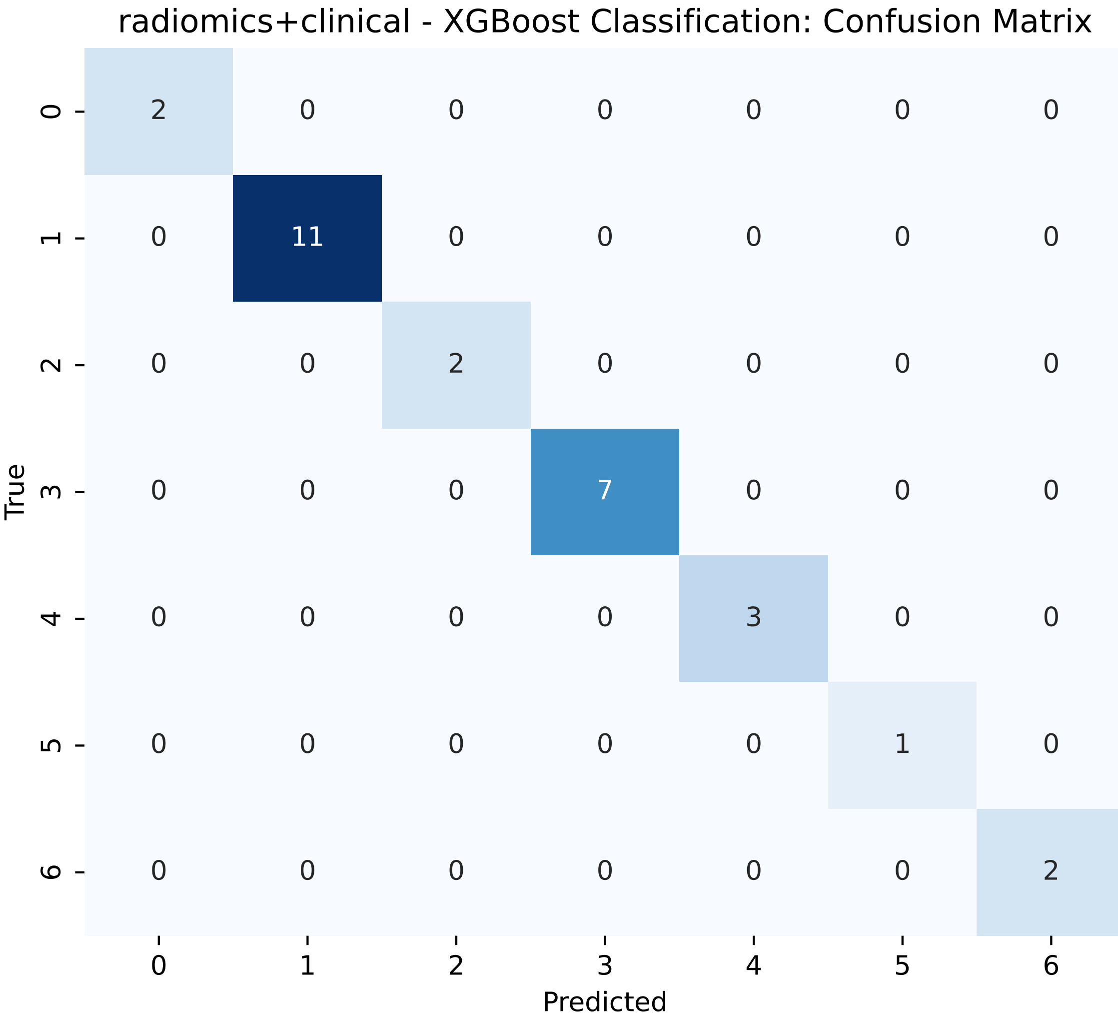

radiomics - LightGBM Classification: Confusion Matrix
True
Predicted
0
1
2
3
4
5
6
2 0 0 0 0 0 0
0 11 0 0 0 0 0
0 0 2 0 0 0 0
0 0 0 7 0 0 0
0 0 0 0 3 0 0
0 0 0 0 0 1 0
0 0 0 0 0 0 2

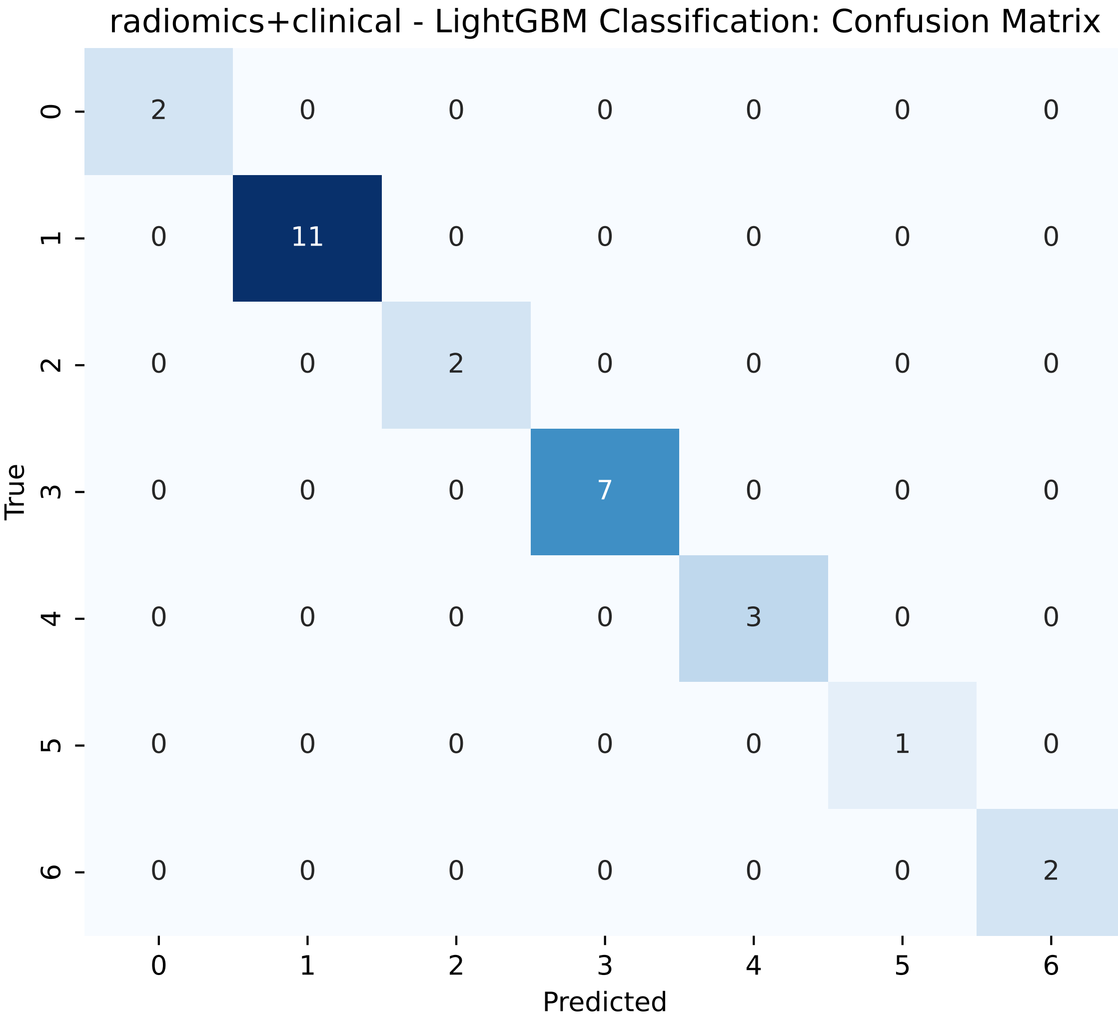

radiomics - Logistic Classification: Confusion Matrix
True
Predicted
0
1
2
3
4
5
6
0 0 0 0 2 0 0
0 0 0 0 10 1 0
0 0 0 0 2 0 0
0 0 0 0 3 4 0
0 0 0 0 3 0 0
0 0 0 0 0 1 0
0 0 0 0 2 0 0

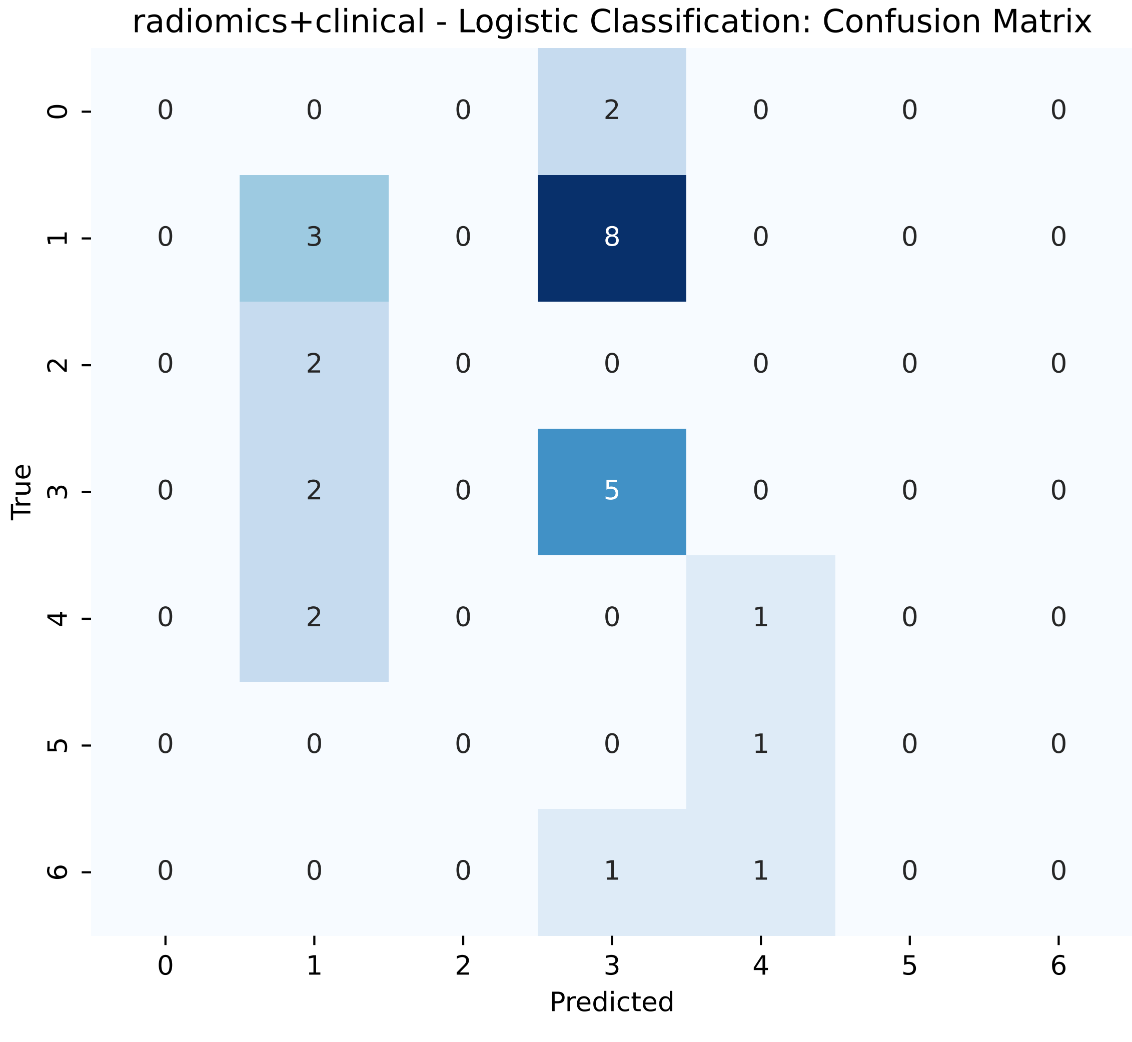

radiomics+clinical - CatBoost Classification: Confusion Matrix
True
Predicted
0
1
2
3
4
5
6
2 0 0 0 0 0 0
0 11 0 0 0 0 0
0 0 2 0 0 0 0
0 0 0 7 0 0 0
0 0 0 0 3 0 0
0 0 0 0 0 1 0
0 0 0 0 0 0 2



radiomics+clinical - RandomForest Classification: Confusion Matrix
True
Predicted
0
1
2
3
4
5
6
2 0 0 0 0 0 0
0 11 0 0 0 0 0
0 0 2 0 0 0 0
0 0 0 7 0 0 0
0 0 0 0 3 0 0
0 0 0 0 0 1 0
0 0 0 0 0 0 2


radiomics+clinical - XGBoost Classification: Confusion Matrix
True
Predicted
0
1
2
3
4
5
6
2
11
2
7
3
1
2
0


radiomics+clinical - LightGBM Classification: Confusion Matrix
True
Predicted
0
1
2
3
4
5
6
2 0 0 0 0 0 0
0 11 0 0 0 0 0
0 0 2 0 0 0 0
0 0 0 7 0 0 0
0 0 0 0 3 0 0
0 0 0 0 0 1 0
0 0 0 0 0 0 2


radiomics+clinical - Logistic Classification: Confusion Matrix
True
Predicted
0 1 2 3 4 5 6
0 0 0 2 0 0 0
0 3 0 8 0 0 0
0 2 0 0 0 0 0
0 2 0 5 0 0 0
0 2 0 0 1 0 0
0 0 0 0 1 0 0
0 0 0 1 1 0 0